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Predicting the compressive strength of concrete incorporating waste powders exposed to elevated temperatures utilizing machine learning

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The addition of powders from waste construction materials as partial cement substitute in concrete represents a significant step toward green concrete construction. High temperatures have a substantial influence on concrete strength, resulting in a reduction in mechanical properties. The prediction of the impacts of waste powders on concrete strength is an important topic in sustainable construction. Such models are needed to understand the complex interactions between waste materials' powders and concrete strength. In this study, three machine learning approaches, extreme gradient boosting (XGBoost), random forest (RF), and M5P, were used for constructing the prediction model for the impact of elevated temperatures on the compressive strength of concrete modified by marble and granite construction waste powders as partial cement replacements in concrete. Dataset of 324 tested cubic specimens with four input variables, waste granite powder dose (GWP), waste marble powder (MWP), temperature (T), and duration (D) were chosen for developing the prediction models. The output was the concrete compressive strength (CS). MWP and GWP ranged between 0 and 9%, temperatures were ranged between 25 °C and 800 °C, and duration up to 2 h. Hyperparameters in the RF and XGB models were optimized using grid search. K-fold cross-validation and several statistical measures, including R^2MAPE, RMSE, and MAE, were utilized to validate and check the accuracy of the proposed models. The developed models were evaluated against experimental data and previously established models. The XGB model demonstrated the highest R^2 of 0.9989, alongside the lowest prediction errors: MAE of 0.1351 MPa, RMSE of 0.1842 MPa, and MAPE of 0.48%. The results showed that the XGB prediction model for the concrete compressive strength outperformed the other proposed models. The SHAP analysis, Individual Conditional Expectation (ICE), and Partial Dependence Plots (PDP) revealed that GWP and MWP positively influence the compressive strength, while the temperature exerts the most negative influence on predicting the compressive strength. Finally, a graphical user interface (GUI) for the compressive strength of concrete containing GWP and MWP subjected to elevated temperatures has been created, which may be of considerable assistance, guidance, and efficiency in research and construction industry contexts.

Keywords Machine learning, Mechanical properties, Waste powders, Compressive strength, Concrete, Elevated temperature

Abbreviations

ANNs	Artificial neural networks
CS	Compressive strength
CSH	Calcium silicate hydrate

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D	Exposure duration
FL	Fuzzy logic
GA	Genetic algorithms
GEP	Gene expression programming
GWP	Granite waste powder
ICE	Individual Conditional Expectation
LR	Linear regression
ML	machine learning
MLR	Multiple linear regression
MWP	Marble waste powder
OPC	Ordinary Portland cement
PDP	Partial Dependence Plots
PPF	Polypropylene fiber
R	Correlation coefficient
R^2	Coefficient of determination
RF	Random forest
RMSE	Root mean squared error
SHAP	SHapley Additive exPlanation
SVM	support vector machines
SVR	support vector regression
T	Temperature
WCA	Water cycle algorithms
XGB	extreme gradient boosting
MAPE	Mean absolute percentage error
MAE	Mean absolute error
LM	Linear model
SD	Standard deviations
SDR	Standard deviation reduction
SHAP	Shapley Additive explanation

It is clear that the construction industry is a significant contributor to the enormous quantities of waste material via its operations. As a result, significant quantities of waste end up in landfills due to construction activities. Therefore, addressing and decreasing the negative impacts of the construction sector on the environment has become an urgent concern[1]. At the core of the construction sector, concrete production emerges as the fundamental and essential building material. Concrete stands as one of the preeminent construction materials globally, distinguished by its advantageous properties such as superior compressive strength (CS), enhanced durability, and remarkable versatility[2]. A substantial proportion, estimated at 46%, of the materials required for infrastructure development consists of cementitious composites, such as mortar and concrete[3]. The manufacturing of concrete involves resource-intensive procedures, utilizing aggregates, cement, and water as its primary constituents. Among these elements, the global cement production rates have been increasing. Cement manufacturing constitutes about 7% of global CO_2 emissions, making it the third-largest source of CO_2 emissions[4]. The inclusion of waste materials as partial substitutions of cement in concrete to reduce the impact of CO_2 emissions without significantly impacting the strength, durability, or cost of the material is a crucial aspect in the construction industry[5–7].

In the present study, two waste materials, an igneous rock composed primarily of silicon dioxide SiO_2, named as granite and a metamorphic rock composed primarily of calcium carbonates $CaCO_3$, named marble, these two materials commonly used in construction processes especially finishing and decoration were selected with the aim of using their powders as partial cement substitutions. These wastes can be disposed of in an economical and sustainable manner. The used marble and granite were obtained from the city of Shiek El Thoaban, located south of Cairo in Egypt[8].

Several studies have investigated the effect of granite and marble powder as partial cement replacements on various properties of concrete. For instance, increased percentages (up to 15%) of granite powder were utilized as a partial substitution material for cement while examining its influence on several properties of concrete and mortar[9]. In the case of mortar, they found that the optimum percentage for improving early-age compressive strength was 7.5% while achieving a 10% replacement ratio resulted in better performance than the reference mixture. Regarding concrete, the most favorable replacement percentage was 5% for enhancing both tensile and compressive strengths. Likewise, the best results in tensile strength were recorded when using granite powder as an additional percentage to cement at 10%. The specimens containing 5% waste marble powder (MWP) demonstrated the highest tensile and compressive strengths, while 10% GWP samples outperformed the control mixture[10]. Ghorbani et al.[11] investigated the adverse exposure conditions' influences on the performance of concrete modified by granite powder. They found that incorporating 10% GWP in concrete exhibited the greatest strength among all variants, while incorporating 20% GWP demonstrated superior acid and chloride resistance.

Corinaldesi et al.[12] concluded that MWP serves as a filler material. Due to its filler action, substituting 5% of the cement with marble powder resulted in 12% improvement in the compressive strength, 5% increase in bending strength value, and a reduction in the porosity. The main highlighted outcome of study conducted by Rana et al.[13] is the feasibility of replacing 10% of Portland cement with marble powder. However, it's worth noting that the extensive surface area of marble powder does have an adverse effect on concrete workability. Nevertheless, it is still possible to achieve concrete with the required strength and improved durability by incorporating 10% marble powder. This is because the finer particles of marble powder, in comparison to cement, occupy voids and compact the concrete matrix. As a result, the compact microstructure enhances concrete's resistance to permeation, chloride migration, and corrosion. Zhang et al.[14] examined the effects utilizing MWP and silica fume as SCM on the strength of concrete. They reported that the splitting tensile strength experiences a notable increase up to a MWP content of 15%, after which it starts to decline when the MWP content reaches 20%. Moreover, 5% MWP or higher proves advantageous for enhancing compressive strength. Rashwan et al.[15] proposed the use of up to 20% marble and granite sludge as a possible substitute for cement in concrete applications. This substitution had no discernible effect on the mechanical or physical properties of the concrete. Talah et al.[16] found that a 15% replacement ratio is ideal for replacing cement with marble powder in high-strength concrete. At this percentage, a significant improvement in several concrete durability properties was observed while maintaining the concrete's compressive strength values without any decline. Pereira et al.[17] investigated the use of marble and granite waste powders in self-compacting concrete production. Their results indicated that incorporating these wastes did not affect the concrete's density. Furthermore, the waste contributed to concrete stabilization and improved certain durability test results. This behavior was likely due to the marble and granite waste acting as a filler, which aided in disconnecting the concrete's pores. Nega et al.[18] investigated the effects of partially replacing cement in mortar with a blend of MWP and GWP. Initial tests identified a 50% MWP to 50% GWP blend as yielding the highest strength. This optimized marble and granite waste powder (MGWP) blend was then incorporated into mortar at varying increments (0–30%). The study found that GWP classifies as a natural pozzolan (Class N). While increasing MGWP content reduced mortar workability, additions up to 15% enhanced bulk density, compressive strength, and homogeneity, with 10% MGWP yielding optimal performance. Microstructural analysis confirmed that MGWP improved C–S–H and C–H bonding, resulting in a denser morphological structure, particularly at 10% MGWP content. Mahmoud et al.[19] investigated the use of waste granite, marble, granodiorite, and ceramic powders as partial cement replacements in normal concrete. They achieved the best compressive strength results with replacement levels of 9% for granite, 7% for ceramic, 7% for granodiorite, and 5% for marble powders. These replacements led to corresponding compressive strength improvements of 25.6%, 24.7%, 23.1%, and 10.3%, respectively.

Due to its composite nature and the varying thermal properties of its constituents, the capacity of concrete's resistance of fire and elevated temperatures is a highly complex attribute[20–22]. Critical temperatures exist at which specific physical and performance alterations in concrete transpire. Concrete exposed to high temperatures experiences a reduction in binding properties[8]. Cracks generally initiate at 600 °C and increase in extent and size at 800 °C[23]. The decomposition of portlandite at approximately 500 °C is a significant chemical change, alongside the dehydration processes. The thermal event significantly contributed to the paste's deterioration. At elevated temperatures of approximately 800 °C, the mass losses observed in thermogravimetric analyses were primarily ascribed to the decomposition of carbonates, including dolomite and calcite[24]. The temperature of 800 °C resulted in approximately 50% reduction in the compressive strength[25]. When concrete is exposed to high temperatures, its compressive strength suffers significantly due to aggregate expansion. When concrete is subjected to high temperatures, the choice of aggregates has a considerable impact on its relative strength. According to research, siliceous aggregates expand the fastest and cause the most damage[8].

Recent advancements in machine learning (ML) techniques have effectively facilitated the prediction of concrete properties across several industries, hence minimizing the time required for testing procedures. The strength of ML models lies in their ability to autonomously learn from data, identifying subtle patterns and complex interdependencies between various factors and the desired target variables, which ultimately leads to enhanced predictive accuracy and robustness[26]. These techniques include M5P[27–33]; random forest (RF)[30,32,34–43] XGBoost[30,44–47]. Many studies have investigated the effects of using hybrid construction waste powders as partial substitutes for cement in concrete. The incorporation of waste materials in concrete enhances performance and results in significant strength properties[8]. This study investigated the influence of four key variables on the residual compressive strength of concrete incorporating granite and marble waste following exposure to elevated temperatures. An extensive experimental program, comprising 324 data points, was conducted to achieve this. The variables under investigation included the type of waste material, its replacement ratio with cement, the exposure temperature (from ambient to 800 °C), and the exposure duration (1–2 h). Concrete incorporating 9% GWP and 5% MWP as partial cement replacements exhibited enhanced compressive and tensile strengths[8]. Limited research has examined the impact of elevated temperatures on compressive strength of concrete containing waste powders. Research into the impact of elevated temperatures on compressive strength of concrete incorporating MWP and GWP under high-temperature conditions is essential.

To evaluate data sensitivity and analyze the influence of input variables on ML model predictions, two powerful and complementary approaches of SHapley Additive exPlanations (SHAP) and Individual Conditional Expectation (ICE) with Partial Dependence Plots (PDP) are commonly used in recent research[48,49]. SHAP analysis quantifies the independent contribution of each feature to the predicted outcome, proving highly effective for interpreting the outputs of complex ML models by providing deep insights into nonlinear patterns and variable interactions, thereby enhancing model transparency and supporting informed decision-decision-making. Furthermore, SHAP analysis offers valuable insights into complex nonlinear and interactive relationships between variables, which facilitates informed decision-making and improves the transparency and accuracy of predictive models[50]. On the other hand, ICE with PDP is a robust method for exploring feature effects in ML. While PDP illustrates the average effect of a feature on model predictions, ICE offers a more detailed view by showing how the feature influences predictions for individual instances[51]. This combined approach is particularly beneficial for identifying heterogeneous effects and understanding how a feature's impact might vary across different data points. Together, these methods offer a comprehensive understanding of feature importance, interactions, and overall model behavior.

Research significance

Through the current study, a wide scale experimental study has been made for the first time to investigate the impact of high temperatures and various exposure times on the compressive strength of waste concrete containing single and combined replacement ratios of granite and marble waste powders for cement. A total of 324 results of compressive strength datapoints were utilized for developing compressive strength prediction models for concrete modified by MWP and GWP under elevated temperatures effect. This study employed ML techniques, including RF, XGB, and M5P. Previous studies focusing on the development of prediction models utilizing various machine learning techniques for concrete behavior, particularly compressive strength after exposure to high temperatures, remain limited. No previous study has focused on developing prediction models that assess the effects of the hybrid incorporation of MWP and GWP as partial cement replacements using the machine learning techniques that are used in this study. Also, this study uses ICE with PDP plots for exploring feature effects on predicting the compressive strength of concrete containing MWP and GWP and being exposed to elevated temperatures. This study seeks to create three machine learning prediction models and efficient predictive equations for estimating the compressive strength of concrete containing two types and replacement ratios of marble and granite waste powders. Also, SHapley Additive exPlanations (SHAP) and Individual Conditional Expectation (ICE) with Partial Dependence Plots (PDP) were utilized to evaluate data sensitivity and analyze the influence of input variables on the compressive strength prediction. Table 1 presents a systematic overview of key studies focused on predicting the compressive strength of concrete.

Experimental data

The utilized experimental data for developing the prediction model was gathered from previous research[8]. The description of the experimental data is described in the following section.

Materials properties

The materials utilized were cement, water, coarse aggregate (CA), fine aggregate (FA), superplasticizer, GWP, and MWP. The cement was CEM I 42.5 N conforming ASTM C 150[68] and has a specific gravity of 3.17, surface area of 3535 cm^2/g, and setting times of 157 and 298 min. The used coarse aggregate was dolomite having a bulk density of 1668 kg/m^3, a specific gravity of 2.67, a nominal maximum size of 19 mm, a crushing value of 21%, and absorption of 0.55%. The fine aggregate used was silica sand having a bulk density of 1728 kg/m^3, a specific gravity of 2.52, a fineness modulus of 2.4, and absorption of 0.85%. The gradations of both CA and FA conforming ASTM C33[69] are shown in Fig. 1. Clean and fresh water was utilized. The used superplasticizer was Sikament NN conforming ASTM C494, type F (ASTM International, 2017). The superplasticizer dose was 0.4% b weight of cementitious materials.

GWP and MWP were utilized as partial cement replacements in concrete (Fig. 2). The MWP has a white color, a specific gravity of 2.64, and a surface area of 7650 cm^2/g. The GWP has a gray color, a specific gravity of 2.72, and a surface area of 14,400 cm^2/g[8]. The gradations of the used powders are illustrated in Fig. 3 and the chemical analysis is shown in Table 2.

Waste granite and marble were obtained from ornamental stone industry factories located in Cairo, Egypt. The chemical analysis results (Table 2) show that GWP has a moderate level of alumina (Al_2O_3) and a substantial presence of SiO_2, demonstrating pozzolanic activity that results in an improvement in the concrete hardened properties[70] as well as an enhancement in the concrete durability through their reaction with produced calcium hydroxide, forming cementitious compounds[24]. The chemical analysis results (Table 2) confirmed that CaO is the predominant component of MWP. The high LOI ratio signifies a substantial presence of volatile components in the sample that are eliminated by heating to 1000 °C. Marble, which is mostly composed of calcium carbonate that decarbonizes and thermally decomposes when heated to temperatures between 650 and 950 °C[71].

Mix proportions

The control mixture of concrete comprises fine aggregate, coarse aggregate, cement, water, and superplasticizer of 735 kg, 1165 kg, 350 kg, 140 kg, and 1.4 kg, respectively for one cubic meter. The cement content was partially replaced with GWP or MWP at replacement levels ranging between 1% and 9% by weight of cement. The hybrid incorporation of GWP and MWP as partial cement substitute was employed with total replacement ratios up to 18% by weight of cement. The mixing was done using pan type concrete mixer. The concrete specimens were cubic specimens of 100 × 100 × 100 mm. The samples were tested under compression after 28 days of curing. Three samples were tested for each concrete mixture and the mean result was recorded. The concrete cubes were subjected to temperatures of (200, 400, 600, and 800 °C) for one and two hours in a furnace conforming with standard ISO 834 fire resistance curve to examine the effect of temperatures on compressive strength of concrete containing GWP and MWP.

Test results

Figure 4 illustrates the effect of utilizing GWP and MWP as individual cement replacement materials on concrete compressive strength of at several temperatures. The findings indicate that GWP significantly improved the compressive strength of concrete across all replacement ratios with cement (Fig. 4a). The trends of the compressive strength were approximately the same at all temperature exposure and duration. The notable improvement in compressive strength can be attributed to the unique characteristics of GWP. The GWP possesses a filling effect and a pozzolanic nature as a result of the amorphous silicates and aluminates that make up about 81% of the GWP composition. The results indicated that MWP gradually enhanced the concrete compressive strength, reaching its optimal substitution level of 5% with cement; beyond this percentage, the compressive strength began to decrease at all temperature exposure and duration. The notable enhancement in strength at a 5% replacement

Reference	Sample size	Study target	Used ML techniques	Input variables	Main results
Dahish and Almutairi (2025)[32]	169 values of compressive strength collected from literature studies	Predicting compressive strength of concrete containing nano alumina and carbon nanotubes subjected to elevated temperatures.	M5P RF	Temperature degree Heating duration Carbon nanotubes dose Nano alumina dose	RF model outperformed M5P model. RF-R^2 0.9965 M5P-R^2 0.9328
Alyami et al. (2024)[45]	348 values of compressive strength collected from literature studies	Predicting compressive strength of concrete containing rice husk ash	RF XGBoost Light GBM Ridge regression	Fine aggregate content Coarse aggregate content w/c ratio Superplasticizer ratio Rice husk ash content	Light GBM demonstrated superior performance Training R values of 0.943 (ridge regression) 0.981 (RF) 0.985 (XGBoost) 0.996 (Light GBM)
Singh et al. (2019)[52]	83 compressive strength values	Predicting compressive strength of high-strength concrete	RF M5P	Cement content Silica fume content Fly ash content aspect ratio Water content Aggregates weights Fiber content Superplasticizer weight CS	RF regression demonstrates superior accuracy in predicting CS compared to the M5P model. RF-R^2 = 0.876 RF-RMSE = 8.014 MPa RF-MAE = 0.255 MPa M5P-R^2 = 0.807 M5P-RMSE = 8.836 MPa M5P-MAE = 0.286 MPa
Zhang et al. (2019)[53]	390 cylindrical specimens, grouped into 130 batches of three specimens each	Predicting the compressive strength of lightweight self-compacting concrete	RF	w/c ratio PPF (%) Steel fiber (%) Scoria content Crumb rubber content Aggregates weights	A high correlation coefficient RF-$R = 0.97$
Nguyen et al. (2020)[54]	1030 specimens from 17 literature studies	Predicting concrete compressive strength.	XGBoost ANN SVM	Cement content Fly ash content Blast furnace slag content Water content Aggregates weights Superplasticizer weight Curing age	The three machine learning models, ANNs, SVMs, and XGBoost, achieved comparable R^2 values around 0.97. XGBoost model demonstrated statistically significant robustness, and predictive accuracy when compared to the ANN and SVM models.
Kocamaz et al. (2021)[55]	40 sets of available data	Predict compressive strength and ultrasonic pulse velocity of concrete.	M5P	Curing age Cement content Fly ash content Silica fume content Blast furnace slag content	The proposed M5P model accurately estimated both CS and UPV values, achieving an R^2 of 0.97
Emad et al. (2022)[56]	274 data sets from literature studies	Predicting compressive strength of ultra-high performance fiber reinforced concrete (UHPFRC).	M5P ANN	Cement content Silica fume content Water content Aggregates weights Superplasticizer weight w/c ratio Fiber characteristics Curing age Curing temperature	The ANN model exhibited superior performance compared to other models, as evidenced by the lowest RMSE and the highest R^2 values.
Li et al. (2022)[57]	10 distinct experimental mixes	Predicting compressive strength of basalt fiber reinforced concrete.	RF BP neural network SVR	Strain Fiber length Confining pressure Volume fraction	RF model demonstrated superior accuracy and a high coefficient of determination (R^2 = 0.96) in predicting the CS of BFRC.
Ahmed et al. (2022)[58]	197 experimental data of HSC mixes with Metakaolin (MK) from literature studies	Predicting compressive strength of high-strength concrete blended with metakaolin.	M5P Linear regression Nonlinear regression Multi-logistic regression	Cement content Curing time w/c ratio MK content Superplasticizer content Aggregates weights SiO_2 content Al_2O_3 content	M5P-tree model demonstrated superior performance, achieving R^2 of 0.77, RMSE of 6.67 MPa, and a scatter index (SI) of 0.096.
Chen et al. (2023)[59]	Collected from 7 literature studies	Predicting compressive strength of high-strength concrete subjected to elevated temperatures.	RF XGBoost Adaptive Boosting	Cement content Nano silica content Fly ash content Silica fume content Water content Aggregates weights Superplasticizer ratio Temperature	RF model obtained the best performance with minimum mean absolute error and root mean square error values and R^2 value of 0.98.
Ibrahim et al. (2023)[60]	425 experimental data of CS of ternary cement concrete from literature studies	Predicting CS of sustainable ternary cement concrete	ANN RF Random Tree Gradient Boosting AdaBoost	Cement content Blast Furnace Slag content Fly Ash content w/c ratio Superplasticizer ratio Aggregates weights	The AdaBoost model demonstrated the best performance. The consistently low absolute and residual errors, coupled with high correlation coefficients, indicate that the developed models are robust and possess sufficient predictive capacity for the CS of ternary cement concrete. This conclusion is further supported by the Taylor diagram
Continued					

Reference	Sample size	Study target	Used ML techniques	Input variables	Main results
Gad et al. (2024)[61]	132 experimental results of (EGC) specimens	Predicting compressive strength of Engineered Geopolymer Composites.	RF XGBoost Cat Boost Decision Trees Extra Trees GBR	Silica fume content Water content Binder ratio Superplasticizer ratio Activator ratio Curing method	GBR, CatBoost, and XGBoost were identified as the most effective models, showcasing exceptional accuracy and generalizability. XGBoost achieved an MAE of 1.2967 MPa and an R^2 of 0.9591.
Harith et al. (2024)[62]	600 experimental results from literature studies	Predicting compressive strength of high-performance self-compacting concrete (HPSCC) from non-destructive tests.	M5P RF Random Tree REP Tree	Rebound number Ultrasonic pulse velocity	The M5P model yielded the most accurate predictions $R = 0.9506$ $RMSE = 4.8452$
Wani & Suthar (2024)[63]	343experimental samples from literature studies	Predicting the CS of Waste Foundry Sand Concrete	ANN RF Random Tree M5P REP tree	Waste Foundry Sand ratio Curing age w/c ratio Aggregate type.	Statistical analysis confirmed the RF model's superior predictive capability in the testing stage, achieving an R value of 0.94, WI of 0.96, MAE of 2.09, RMSE of 2.79, RAE of 30.89%, NSE of 0.88, and RRSE of 33.89%.
Wani & Suthar (2024) [64]	154 observations were gathered from literature studies	Predicting the CS of sustainable natural fiber reinforced polymer concrete	RF XGBoost AdaBoost	Diameter of concrete specimen Unconfined CS of concrete specimen Tensile strength of confining fiber in MPa Specimen type Thickness of confining fiber	The XGBoost model demonstrated superior performance over other models during the testing stage, achieving an R^2 of 0.85, RMSE of 5.05, MAE of 3.83, MSE of 25.48, WI of 0.96, and NSE of 0.85. SHAP analysis further revealed that the unconfined CS of the concrete specimen was the most influential factor in forecasting the CS values.
Ansari et al. (2024)[65]	84 available CS results collected from published literature	Forecasting Compressive Strength of SCM Concrete Incorporating blast furnace slag via ML models.	ANN SVM DT	Blast furnace slag to binder ratio Age Ultrasonic pulse velocity	The ANN model demonstrated superior performance over other ML models in predicting the compressive strength of SCM concrete containing BFS.
Ansari et al. (2024)[66]	96 available CS results taken from the published literature	Predicting the CS of silica fume-modified lightweight concrete subjected to thermal loads	ANN RF AdaBoost	Cement content Silica fume content lightweight pumice aggregate content w/c ratio Superplasticizer ratio Temperature	The AdaBoost model outperformed both the ANN and RF models in terms of prediction accuracy. This superiority was indicated by its consistently lower MSE, RMSE, and MAE values for both the training and testing datasets. Separately, the analysis revealed that temperature was the most influential feature impacting residual compressive strength, with elevated temperatures typically leading to reduced strength
Yu et al. (2025)[67]	350 experimental samples from literature studies	Predicting compressive strength of ultra-high-performance concrete (UHPC) after exposure to varying elevated temperature regimes	RF XGBoost BP neural network Categorical boosting Decision tree Gaussian regression	Temperature degree Heating rate Fiber dosage Aspect ratio Fly ash content Slag content Aspect ratio of steel fibers	The categorical boosting model achieved the highest accuracy, as evidenced by an R^2 of 0.930, an RMSE of 0.097, and an MAE of 0.075 when evaluated on the testing data

Table 1. Summary of key existing compressive strength prediction models.

ratio can be attributed to the fineness of MWP, which compensates for voids among concrete components and enhances microstructural density (Fig. 4b)[8]. The optimal ratios for substituting cement with granite or marble waste powders, as determined in this study, closely align with findings from several prior investigations[9,11–13,72].

The results demonstrate the substantial influence of elevated temperatures exposure on the concrete compressive strength (Fig. 4). Low temperatures up to 400 °C had a negligible effect on strength, with a slight enhancement noted at 200 °C. However, exposure to 800 °C for one and two hours led to a compressive strength decline. The individual use of the waste materials led to an enhancement in concrete compressive strength across different temperatures and exposure durations, with improvement ratios differing based on proportion and type of utilized materials. The results for GWP indicated enhanced compressive strength with increasing cement replacement ratios. MWP demonstrated comparable behavior at standard temperature conditions, with an optimal 5% cement replacement ratio.

The hybrid incorporation of GWP and MWP on compressive strength of concrete containing GWP and MWP at ambient temperature is illustrated in Fig. 5. The compressive strength increased as GWP dose increased up to 9% while the presence of MWP improved the strength for doses up to 5% the after 5% the compressive strength decreased[8].

Figures 6 and 7 show the effect of hybrid use of GWP and MWP on compressive strength of concrete exposed to various temperatures for 1 and 2 h, respectively. A slight strength reduction was observed for mixtures containing both waste materials. This suggests that the distinct mechanisms by which each material enhances concrete compressive strength contribute to overall performance improvements. MWP serves as a void-filling

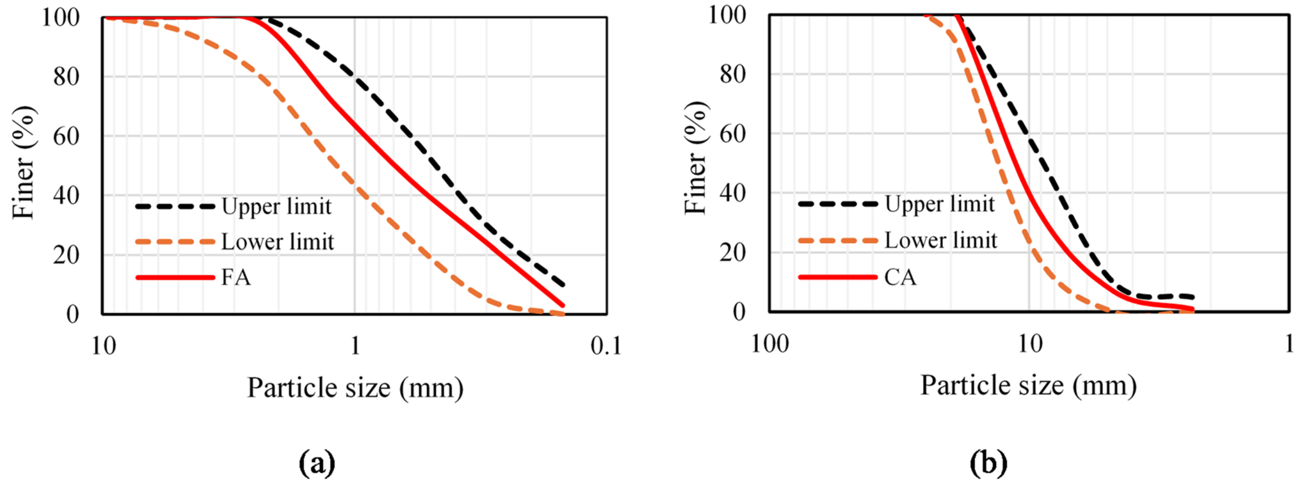


Fig. 1. Aggregates gradations: (**a**) FA, (**b**) CA.

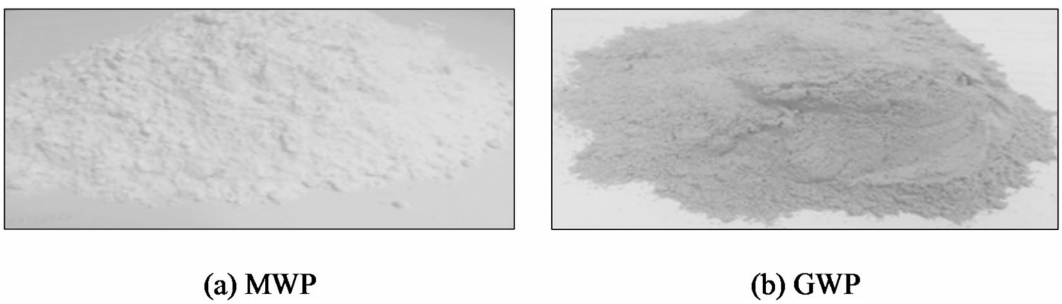


Fig. 2. Powders. (**a**) Marble waste powder, (**b**) granite waste powder[8].

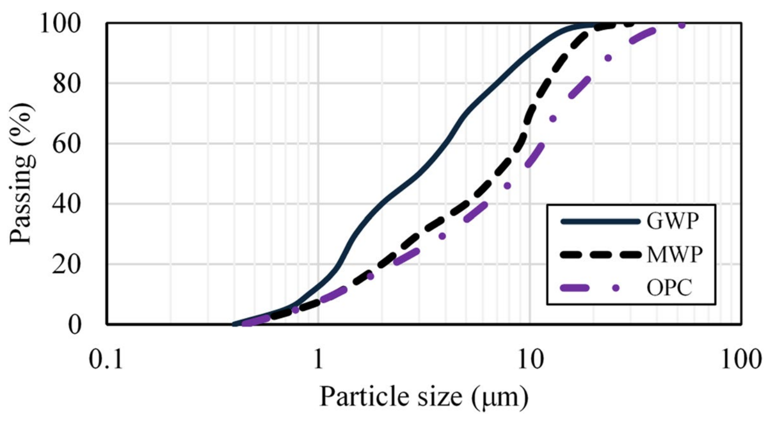


Fig. 3. Particle size distribution of the GWP, MWP, and OPC[8].

Powder	SiO_2	CaO	Al_2O_3	Fe_2O_3	MgO	K_2O	Na_2O	SO_3	P_2O_5	Cl	TiO_2	LOI*
OPC	19.82	64.72	3.1	1.92	2.43	0.72	0.42	2.92	0.19	0.12	0.38	3.11
GWP	67.4	2.94	13.5	3.42	0.8	6.11	3.23	0.08	0.3	0.04	0.49	1.38
MWP	0.49	56.5	0.11	0.03	0.19	0.02	0.01	0.02	---	0.02	---	42.5

Table 2. Chemical analysis of powders[8].

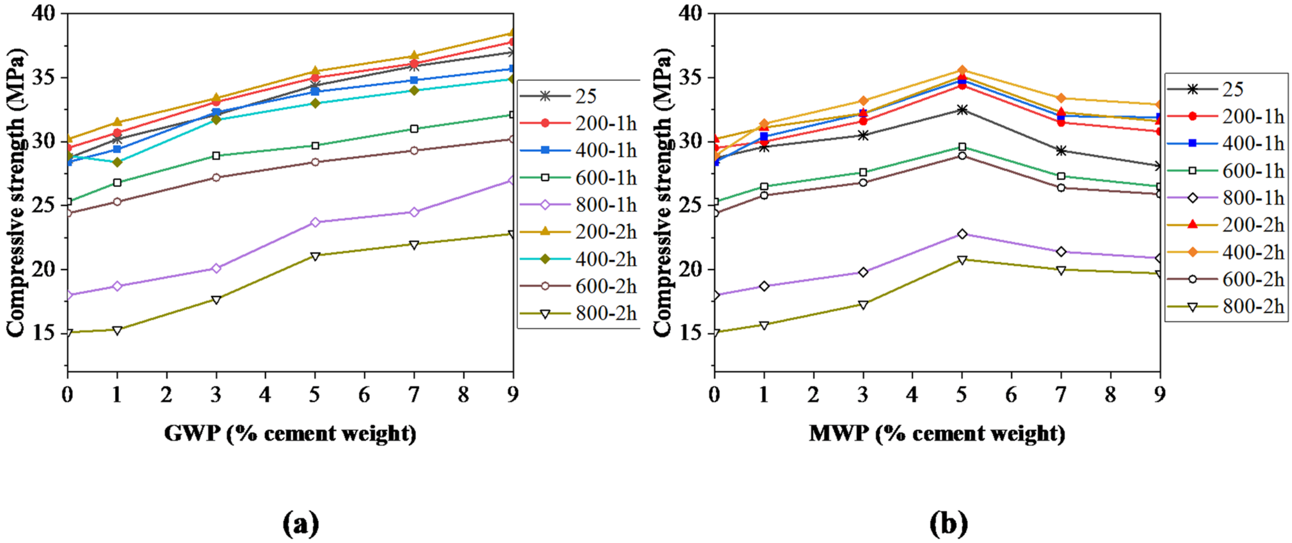


Fig. 4. Effect of individual of cement replacement with GWP or MWP on concrete compressive strength (**a**) cement replacement with GWP, (**b**) cement replacement with MWP.

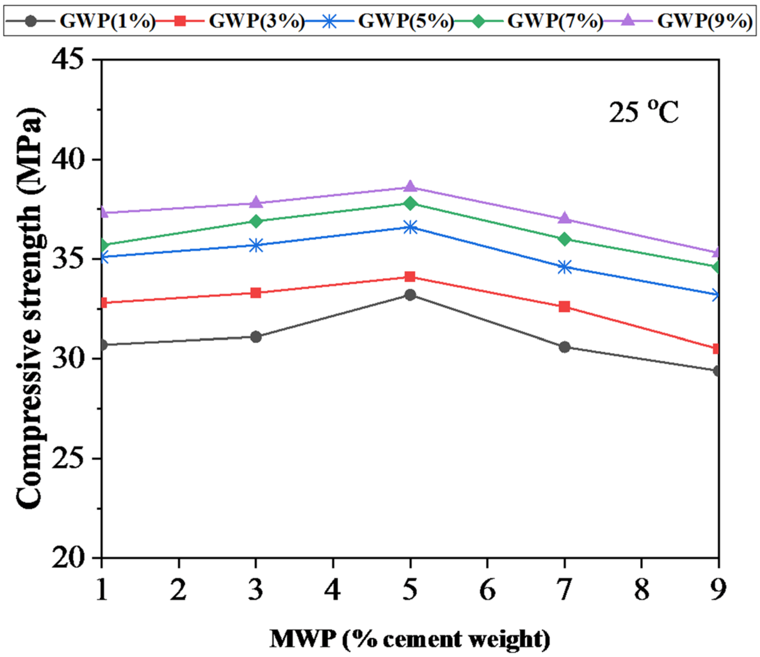


Fig. 5. Impact of GWP and MWP on the room temperature compressive strength.

agent, enhancing the microstructure of concrete, whereas GWP functions as a pozzolanic material, facilitating hydration reactions and increasing the concrete's strength. This has produced a dual effect, yielding superior resistance in all specimens compared to the separate use of each material. The optimal replacement ratio for the combined use of both materials is determined by the sum of their individual improvement percentages, specifically 9% for GWP and 5% for MWP. Materials exhibiting pozzolanic activity, such as GWP, along with fine fillers like MWP, can enhance the concrete resistance against the adverse impacts of high temperatures[73].

The experimental results with varying replacement ratio of cement with GWP and MWP, temperature degree, and exposure duration, were utilized for constructing the compressive strength prediction models for concrete comprises GWP and MWP as partial cement substitutes under elevated temperatures exposure.

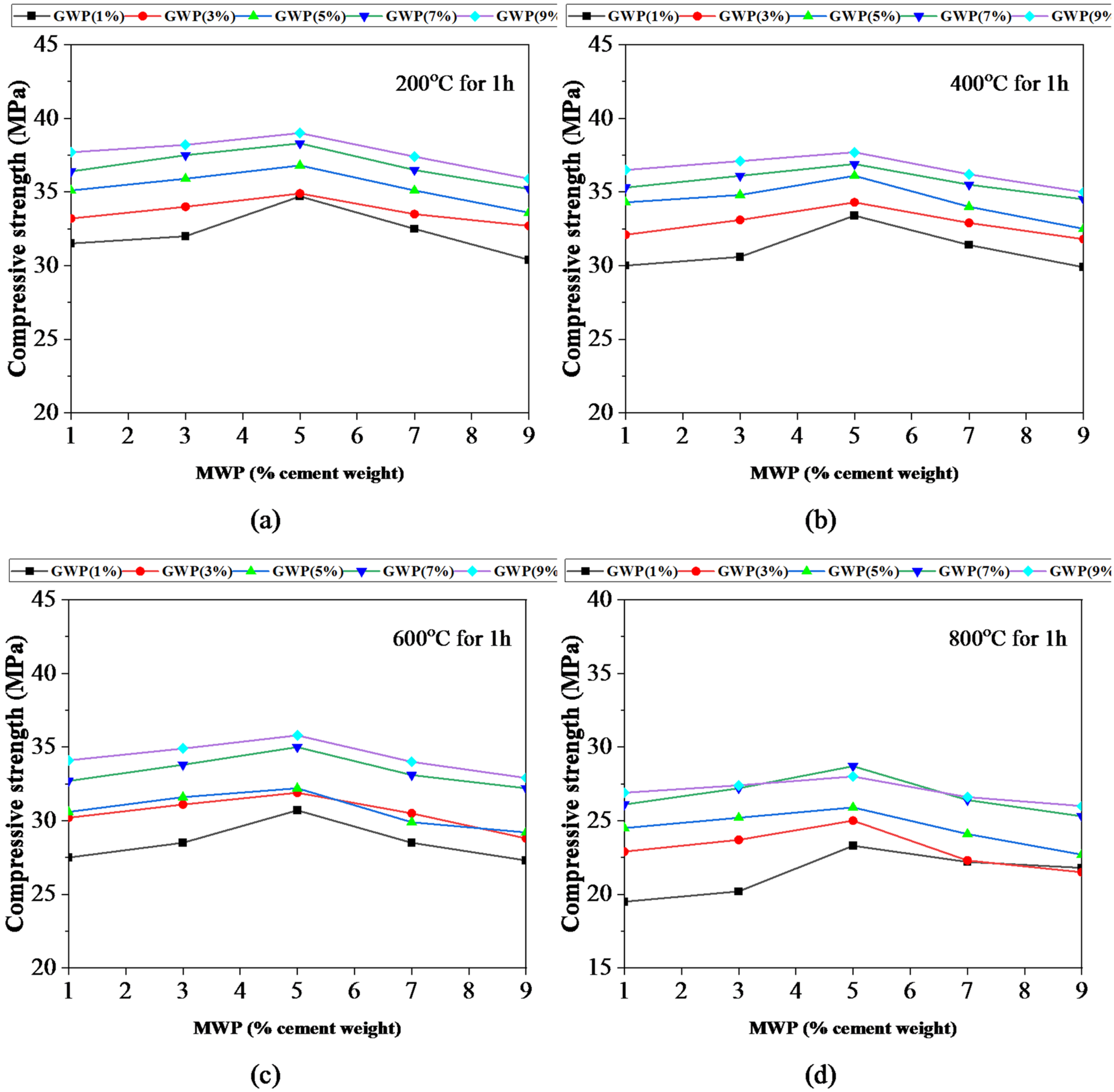


Fig. 6. Impact of GWP and MWP on the compressive strength exposed to temperatures for 1 h: (**a**) 200 °C; (**b**) 400 °C; (**c**) 600 °C; (**d**) 800 °C.

Data and methodology
Experimental data
The developed ML prediction models in the present study were based on the data points of 324 compressive strength results for concrete incorporating GWP and MWP under the effect of elevated temperatures[8]. The input variables were MWP, GWP, temperature degree (T), and duration (D), while the response was the concrete compressive strength (CS). Traditional machine learning begins with raw data, which undergoes feature extraction and data processing as part of the overall data preparation phase[74]. The dataset was divided into a train set for developing the model and a test set for validating the model, adhering to an 80:20 ratio of the entire dataset. The utilized ML approaches were XGB M5P, and RF for developing the models. Table 3 illustrates the dataset statistics. The ranges of the used variables were: GWP dose ranged between 0 and 9%, MWP dose ranged between 0 and 9%, temperature degrees ranged between 25 and 800 °C, and the exposure duration ranged between 0 and 2 h. The range of the output (CS) was 15.1 to 39.4 MPa. The kurtosis and skewness values fell within the acceptable ranges of (-10 to + 10) and (-3 to + 3), respectively[75].

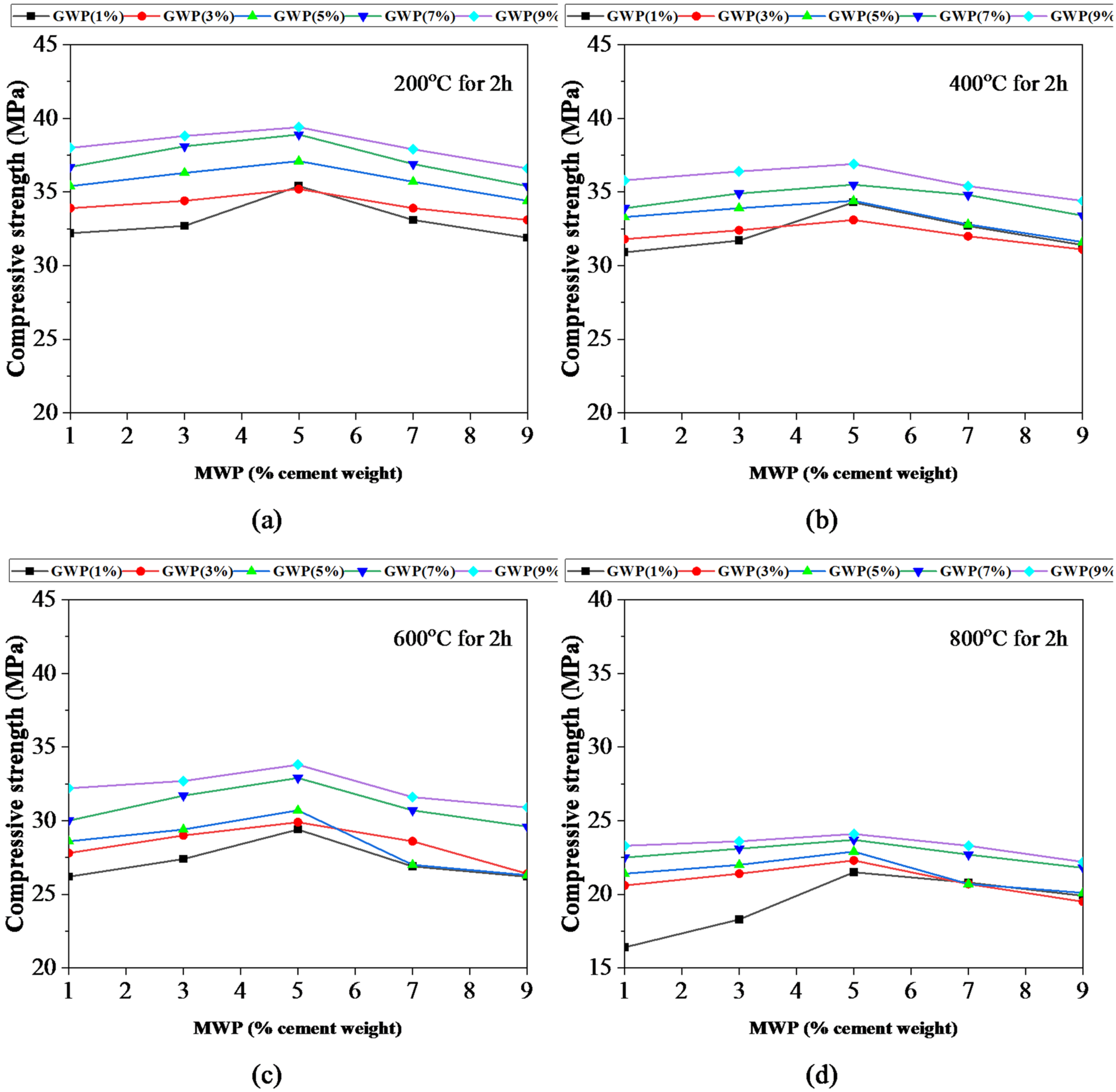


Fig. 7. Impact of GWP and MWP on the compressive strength exposed to temperatures for 2 h: (**a**) 200 °C; (**b**) 400 °C; (**c**) 600 °C; (**d**) 800 °C.

Parameter	Min	Max	Mean	SD	Skewness	Kurtosis
Marble waste powder (MWP)	0	9	4.167	3.1891	0.158	-1.358
Granite waste powder (GWP)	0	9	4.167	3.1891	0.158	-1.358
Temperature (T)	25	800	447.2	258.72	-0.070	-1.220
Duration (D)	0	2	1.333	0.6677	-0.502	-0.743
Compressive strength (CS)	15.1	39.4	30.37	5.4387	-0.705	-0.289

Table 3. Dataset statistics.

Correlation coefficient

The relationships between the output variable (CS) and the input variables (GWP, MWP, T, and D) were assessed through the application of Pearson's correlation coefficient (R). Figure 8a shows the heat map of R. The distribution and interaction of the independent characteristics influencing the output (CS) are fully understood by this statistical study. On a scale from − 1 to + 1, R values indicate strong correlations with ±1 and no correlation with zero. When two variables are positively correlated, it means they are directly related; when they are negatively correlated, it means the opposite is true[76]. In the first row of the graph, you can see how the input variables are related to the output compressive strength. R value of 0.39 indicates a positive correlation between GWP dosage and compressive strength. Both T and D have negative correlations with compressive strength ($R = -0.78$ and − 0.21). In a minor way, MWP and compressive strength are positively related. To better comprehend the interplay between the input and output parameters, Fig. 8b shows the matrix plots. Here, the concrete compressive strength is displayed, along with a series of scatterplots illustrating the intercorrelations between the output parameter and the input variables. This matrix's diagonal plots—histograms—illustrate the dataset's frequency distribution for each variable. All values showed an increase in concrete compressive strength with increasing GWP dose. Up to a 5% MWP dose, the compressive strength of concrete increased; however, compressive strength decreased at doses greater than 5%. As the temperature of the concrete rose, its compressive strength decreased. The compressive strength of concrete dropped as the spent time for heat exposure rose.

Furthermore, multicollinearity concerns might arise in machine learning models. Multicollinearity is a statistical phenomenon in which two or more factors in a regression model are highly connected. The variation inflation factor (VIF) is used to assess multicollinearity in the dataset[77]. The mathematical expression for VIF is shown in Eq. (1).

$$VIF_m = \frac{1}{(1 - R_m^2)} \tag{1}$$

where R_m^2 is the determination coefficient for the x_m variable on the rest of the variables. There is no specific cut-off value for VIF to detect multicollinearity; however, VIF values more than 10 demonstrate the presence of multicollinearity. The VIF values for GWP, WMP, T, and D were 1, 1.2, and 1.2, respectively. The correlation and VIF values are below the cut-off threshold, indicating that there is less likelihood of multicollinearity in the collected data.

Methodology for ML models

This investigation employed RF, M5P-tree, and XGB for predicting and comparing compressive strength of concrete incorporating GWP and/or MWP when subjected to elevated temperatures. A data series of 324 compressive strength results was acquired from previous research[8] to serve as the foundation for the proposed models. Min-max normalization is utilized to guarantee that each feature contributes equally to the training process. Equation (2) presents the mathematical formula for feature normalization.

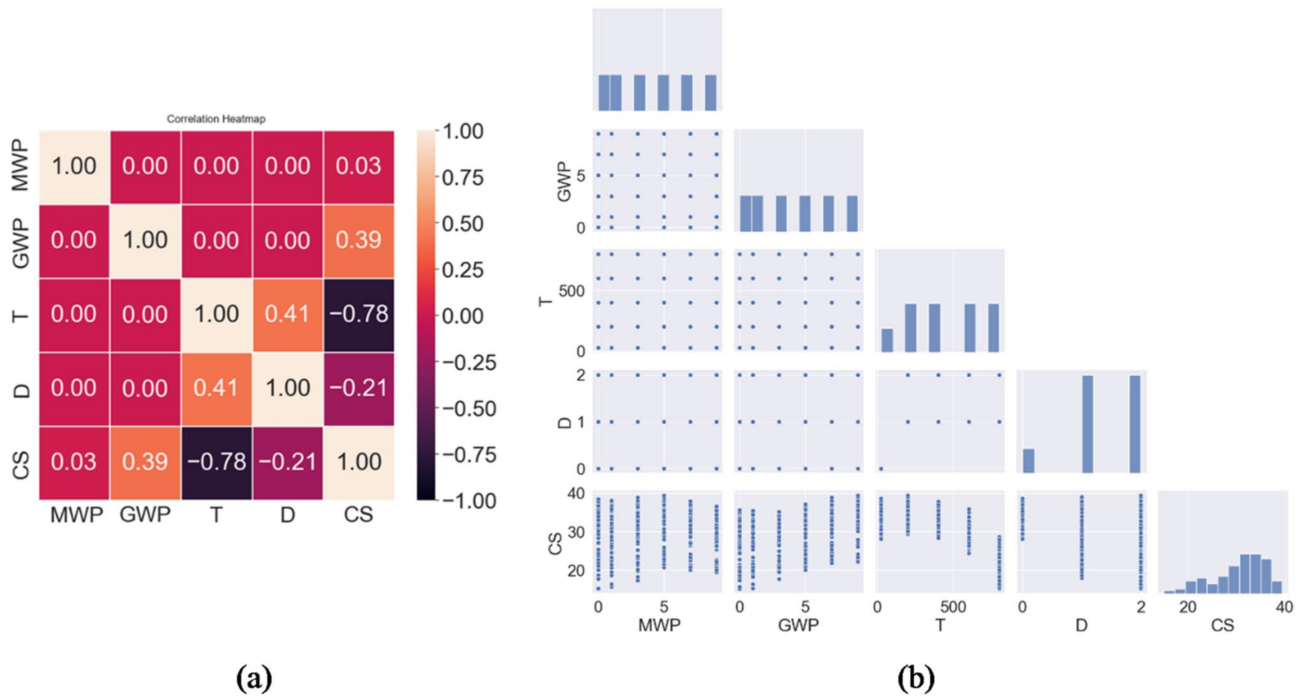


Fig. 8. (**a**) Heatmap of correlation coefficients, (**b**) pairwise matrix plots.

$$X_j = \frac{X_j - X_{min}}{X_{max} - X_{min}} \ (j = 1,2,\ldots,n) \tag{2}$$

where X_{max} and X_{min} are the maximum and minimum values of feature X.

The proposed models' effectiveness was assessed utilizing MAPE, RMSE, MAE, and R^2. To ensure the accuracy of the proposed models, K-fold validation was employed. Figure 9 presents the utilized methodology.

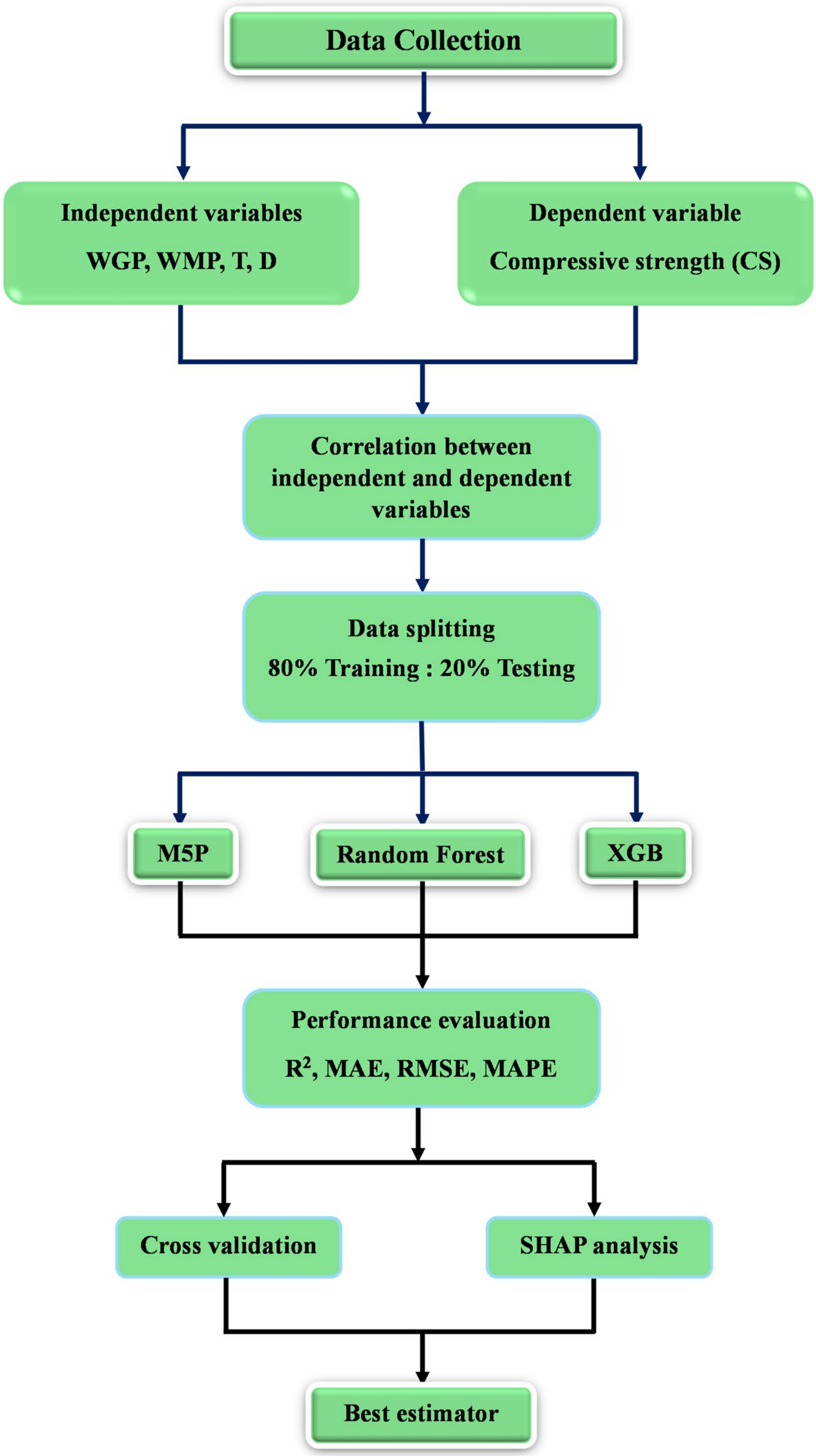


Fig. 9. Adopted research methodology flowchart.

To ensure a thorough examination of the model's generalizability, the dataset was divided into training (80%) and testing (20%) sets using stratified random sampling.

Random forest (RF)

The variables % MWP, %GWP, T, and D, were the determinants of the compressive strength predictions. The RF regressor functions as an ensemble learning approach that utilizes multiple decision trees created during the training process. This approach culminates in an accurate ensemble regressor output by integrating predictions from the various decision trees[36]. The system efficiently handles incomplete and noisy data, leading to reliable outcomes. The reliability of the RF technique is derived from its capability for training the model using instances observed in nature[78]. RF mitigates the overfitting challenges inherent in decision trees by enhancing the accuracy of prediction and optimizing dataset management. The primary RF advantage lies in its implementation of strategies of collective prediction, which effectively minimize errors. Figure 10a presents a depiction of the structure of an RF approach. The result is established by consolidating the forecasts from all participating decision trees. The precision of the model enhances with an increase in the number of trees[78,79].

M5P

The M5P algorithm[80] is the upgraded form of Quinlan's original M5 algorithm[81]featuring a decision tree with leaf nodes LR function. M5P combines the simple structure of decision trees with the efficacy of LR to improve understanding while maintaining prediction precision. This improved method improves the ability to tackle complex problems by breaking them down into smaller parts and then synthesizing responses[32]. This provides thorough details for each LR model component constructed to examine the data set's nonlinear relationships. Figure 10b displays a decision tree structure using with data splitting by selecting input factors that considerably decrease the error at nodes utilizing the SDR at each node. The tree splitting in M5P generates a considerable tree-like configuration, resulting in overfitting. The next stage involves pruning the substantial tree and reconstructing the trimmed subtrees using LR. The substantial tree is pruned in the following phase, and the cut subtrees are recreated using LR functions. The terminal nodes include LR functions (Fig. 10b).

Extreme gradient boosting (XGB)

XGB is a prevalent gradient-boosting algorithm known for its superior performance in classification and regression analyses. In order to improve the predicted accuracy of the original gradient boosting method, XGB incorporates weak models with various training methodologies[82]. XGBoost enhances predictions substantially by integrating the results of many decision trees operating simultaneously, with the efficacy of one tree augmented by reducing the errors of preceding trees[83].

A predictive model with high accuracy and generalizability is developed using the XGB algorithm, which integrates the advantages of multiple ML techniques, such as gradient boosting and decision trees. Decision trees and other weak learners are employed to iteratively improve the ensemble by adding new trees that address the errors made by the previous ones. In comparison to alternative techniques, it exhibits enhanced capability in handling multiple features, operates efficiently with high-dimensional data, and reduces overfitting through the inclusion of a regularization term[84]while also improving the loss function via the second-order Taylor series expansion. This approach enables precise management of model parameters, thereby improving performance and allowing for high levels of adjustability[85].

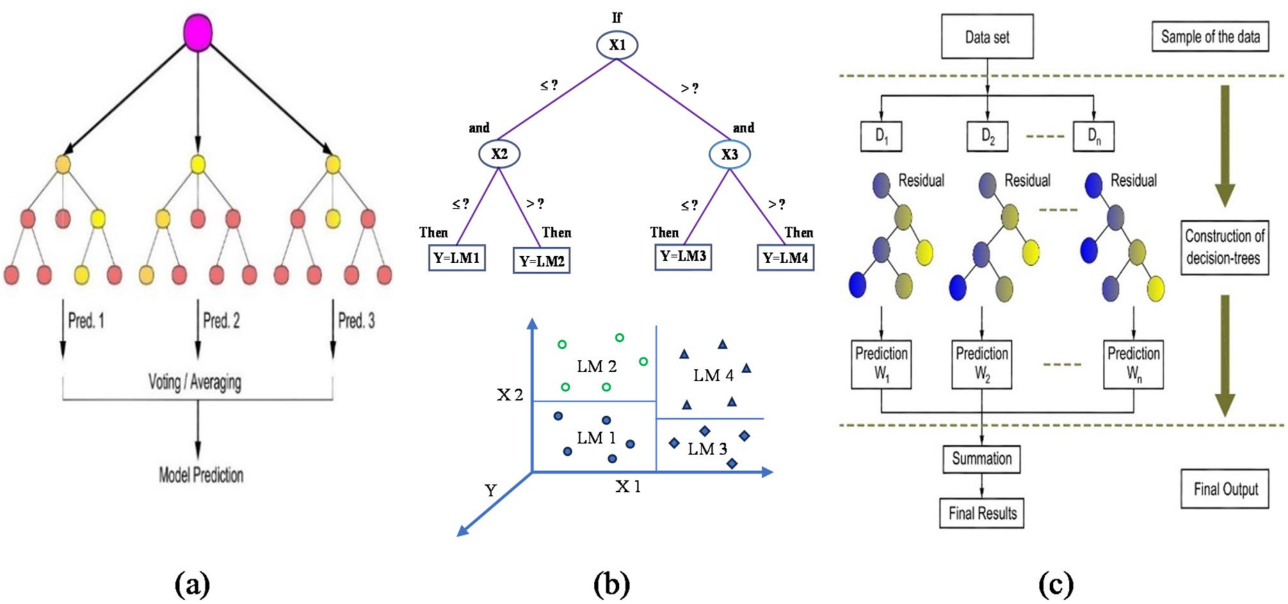


Fig. 10. ML algorithms: (**a**) RF algorithm; (**b**) M5P; (**c**) XGB algorithm.

The objective function consists of the sum of the loss function and the regularization term. The regularization term applies a penalty to complex models, while the loss function measures the difference between the model predictions and actual values. Figure 10c depicts the methodology employed for XGB predictions. The model begins with one decision tree. Subsequently, determine the difference between the original and estimated values. The calculation of the residual constitutes the first step in training a new decision tree to address the shortcomings of its predecessor. The estimated values were updated by incorporating estimates from all trees in the ensemble, and an additional tree was added to the ensemble. The stopping criteria were ultimately met by iterating through the phases of the new tree ensemble and performing residual calculations.

Hyperparameter optimization

The optimization of hyperparameters is crucial in machine learning model training as it enhances the generalizability and robustness of prediction, while reducing the risks of overfitting and underfitting. This study utilized grid search for the optimization of hyperparameters to mitigate overfitting and improve predictive accuracy by examining all potential combinations of hyperparameters to determine the optimal configuration. The evaluation of the models' performance was based on RMSE value. The optimized hyperparameters are illustrated in Table 4.

Model efficiency

The constructed models were assessed utilizing various performance statistical metrics, including root mean square error (RMSE), mean absolute percentage error (MAPE), (mean absolute error) MAE, and R2[86,87]. MAE represents a measure of the difference between predicted and experimental measures. MAE is one of the most often used metrics for calculating error in statistical and machine learning models. Equation (3) represents the mathematical expression for MAE. RMSE is a widely utilized metric for assessing the accuracy of predictive models. RMSE measures the discrepancy between expected values and actual values. A lower RMSE indicates a superior model fit to the data, as we aim for minimal discrepancies between projections and actual outcomes. The mathematical expression for RMSE is illustrated in Eq. (4). MAPE is a widely utilized metric for assessing model accuracy in predictions. It measures the average absolute % errors of actual and predicted values as depicted in Eq. (5). Lower values of MAPE, MAE, and RMSE, indicate enhanced performance of model, reflecting predictions that closely match actual values. A greater R^2 signifies that the dependent variable accounts for a substantial percentage of the variance in the independent variable. Equation (6) presents the expression for R^2 with values range from 0 to 1, serving as an indicator of the efficiency of model fit[88,89].

$$\mathrm{MAE} = \frac{1}{\mathrm{m}} \sum{}_{\mathrm{i}=1}^{\mathrm{m}} |(\mathrm{x_i} - \mathrm{y_i})| \tag{3}$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum{}_{\mathrm{i}=1}^{\mathrm{m}} (\mathrm{x_i} - \mathrm{y_i})^2}{\mathrm{n}}} \tag{4}$$

$$\mathrm{MAPE} = \frac{1}{\mathrm{m}} \sum{}_{\mathrm{i}=1}^{\mathrm{m}} \frac{?\mathrm{x_i} - \mathrm{y_i}?}{\mathrm{x_i}} \tag{5}$$

$$\mathrm{R}^2 = \left[\frac{\sum{}_{\mathrm{i}=1}^{\mathrm{m}} (\mathrm{x_i} - \widehat{\mathrm{x}})(\mathrm{y_i} - \widehat{y})}{\left(\sum{}_{\mathrm{i}=1}^{\mathrm{m}} (\mathrm{x_i} - \widehat{x})^2\right)\left(\sum{}_{\mathrm{i}=1}^{\mathrm{m}} (\mathrm{y_i} - \widehat{y})^2\right)} \right]^2 \tag{6}$$

where, x_i and y_i are the actual and predicted values of CS, $\widehat{x}$ and $\widehat{y}$ are the means of experimental and predicted, respectively, and m is the number of data points.

ICE with PDP

ICE with PDP is an effective approach for investigating feature impacts in ML. PDP depicts how a single characteristic influences predictions on average, but ICE provides a more detailed insight by demonstrating how that feature affects predictions for individual data points[51]. Together, they provide a thorough understanding of feature relationships and model behavior[90].

Algorithm	Hyperparameter	Range	Optimized Value
RF	n_estimators	[10, 200]	23
	max_depth	[10, 30]	12
	max_features	[0.1, 1.0]	0.45
	min_samples_split	[2, 20]	2
	'min_samples_leaf	[1, 10]	1
XGB	n_estimators	[10, 600]	100
	max_depth	[10, 30]	28
	learning_rate	[0.01, 0.3]	0.10
	Subsample	[0.1, 1.0]	0.69

Table 4. Optimized hyperparameters.

Results and discussion
M5P-model
The developed compressive strength model utilizing M5P technique is shown in Fig. 11 (marked LM1 through LM17). The model's general shape is shown in Eq. (7), where a, b, c, d, and e are linear regression constants. Figure 11 indicates the tree-shaped branch relationship of the model. Equation (7) parameters are summarized in Table 5.

$$CS = a^*\,(MWP) + b^*\,(GWP) + c^*\,(T) + d^*\,(D) + e \tag{7}$$

Figure 12 presents the statistical metrics for all the constructed models. The M5P prediction model has superior accuracy, with R^2 of 0.9517 for the train set while the test set has R^2 of 0.9334. The MAE values were 0.9918 and 1.1337 MPa for the train and test datasets, respectively. The RMSE of training and testing datasets were 1.2348 and 1.4087 MPa, respectively. The MAPE values were 3.45% and 4.01% for the train and test datasets, respectively. The dataset exhibits ± 20% margin of error in predicting concrete compressive strength. The findings demonstrate that all observed values fell in the ± 20% margin of errors (Fig. 13a). The scatter plot of the training dataset exhibits a tight grouping of points around the optimal line, indicating minimal variation. The scatter plot of the test dataset reveals that points congregate near the optimal line, however with a marginally greater dispersion compared to the training data.

Random forest
The RF model demonstrates precise output predictions, characterized by a minimal error margin between predictions and actual values, alongside high R^2 values of 0.9961 for the train set and 0.9765 for the test set. The MAE was 0.2919 MPa for the train data and 0.6779 MPa for the test data. The values of RMSE were 0.3648 and 0.8403 MPa for train and test sets, demonstrating that the RF model possesses enhanced accuracy (Fig. 12). The MAPE values were 1.04% and 2.36% for train and test sets. Figure 13b displays scatter plots that compare predicted and actual compressive strength values. The dataset demonstrates a ± 10% margin in compressive

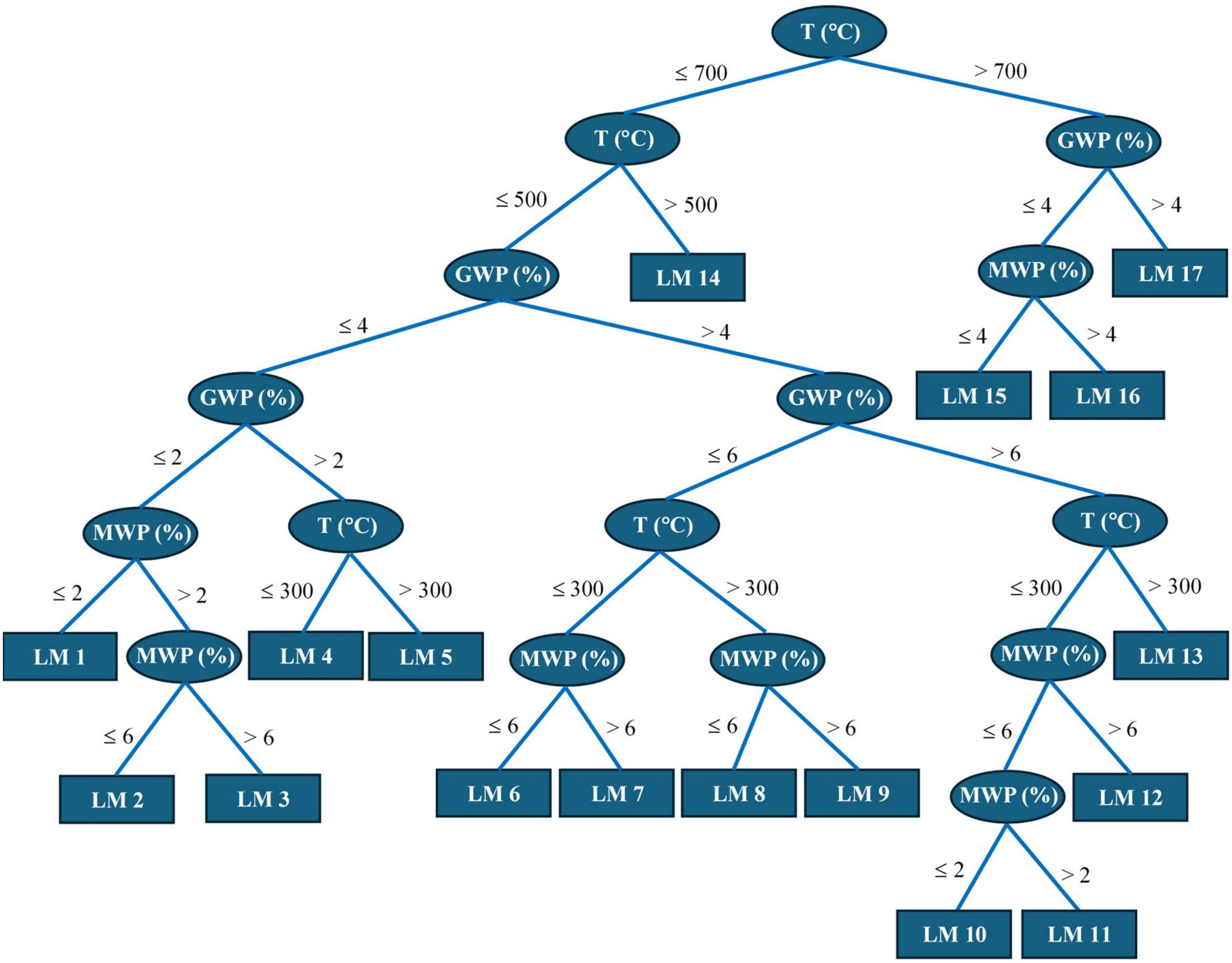


Fig. 11. M5P model with data split diagram.

LM num	Model parameters				
	a	b	c	d	e
1	0.464	0.496	− 0.0036	0.852	30.205
2	0.31	0.247	− 0.0022	0.918	30.164
3	− 0.138	0.247	− 0.0022	1.0471	31.899
4	0.0244	0.2993	− 0.003	0.6944	32.014
5	0.024	0.299	− 0.0032	0.3691	32.22
6	− 0.0013	0.313	− 0.0046	0.363	34.06
7	− 0.101	0.313	− 0.0046	0.326	34.51
8	− 0.025	0.313	− 0.0048	0.0803	34.18
9	− 0.084	0.313	− 0.0048	0.1335	34.33
10	0.0048	0.593	− 0.0041	0.5846	32.47
11	0.05	0.55	− 0.0041	0.5861	32.784
12	− 0.191	0.553	− 0.0041	0.49	33.902
13	− 0.033	0.564	− 0.0045	0.066	32.825
14	0.0	0.674	− 0.0026	− 1.203	30.18
15	0.40	0.888	− 0.003	− 1.951	23.386
16	− 0.0304	0.582	− 0.003	− 1.675	25.72
17	0.038	0.611	− 0.003	− 2.563	26.32

Table 5. Model parameters for compressive strength using M5P-Model (Eq. (7)).

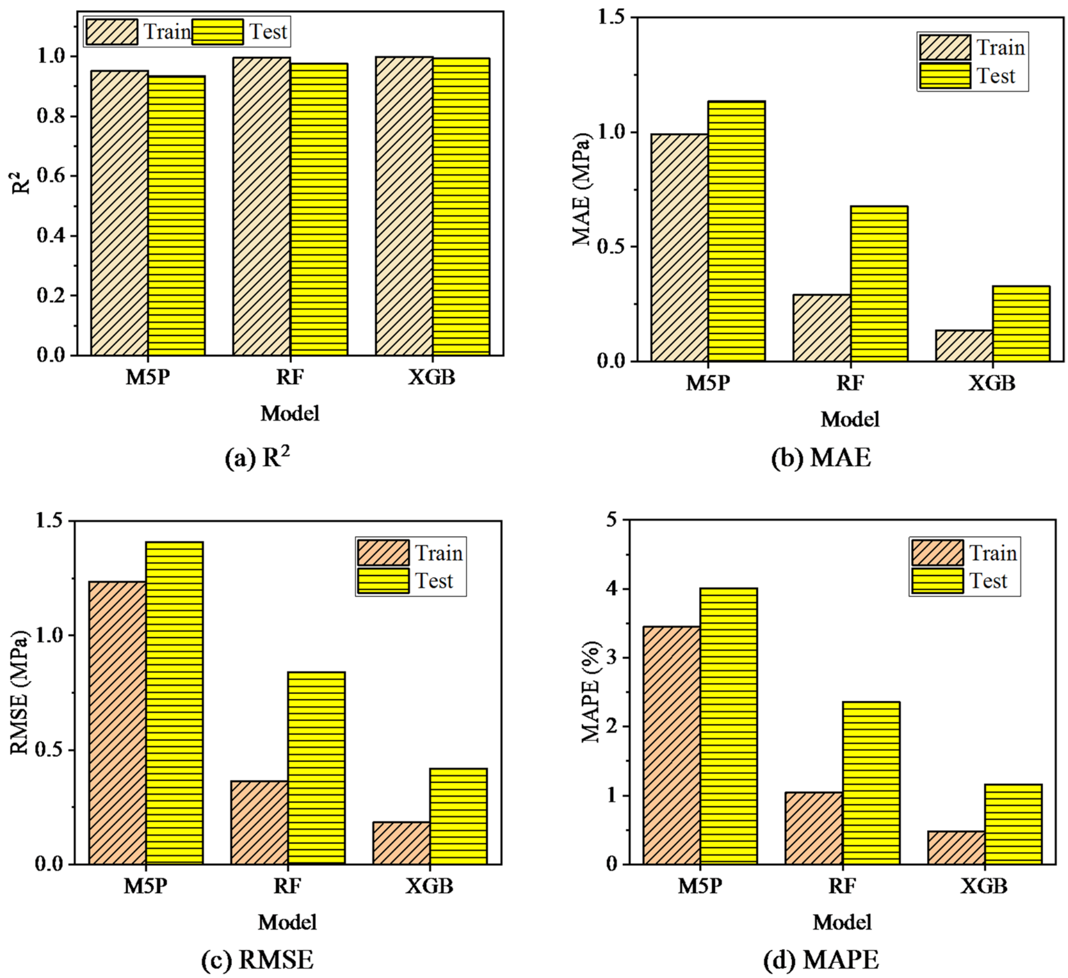


Fig. 12. Performance of the developed models.

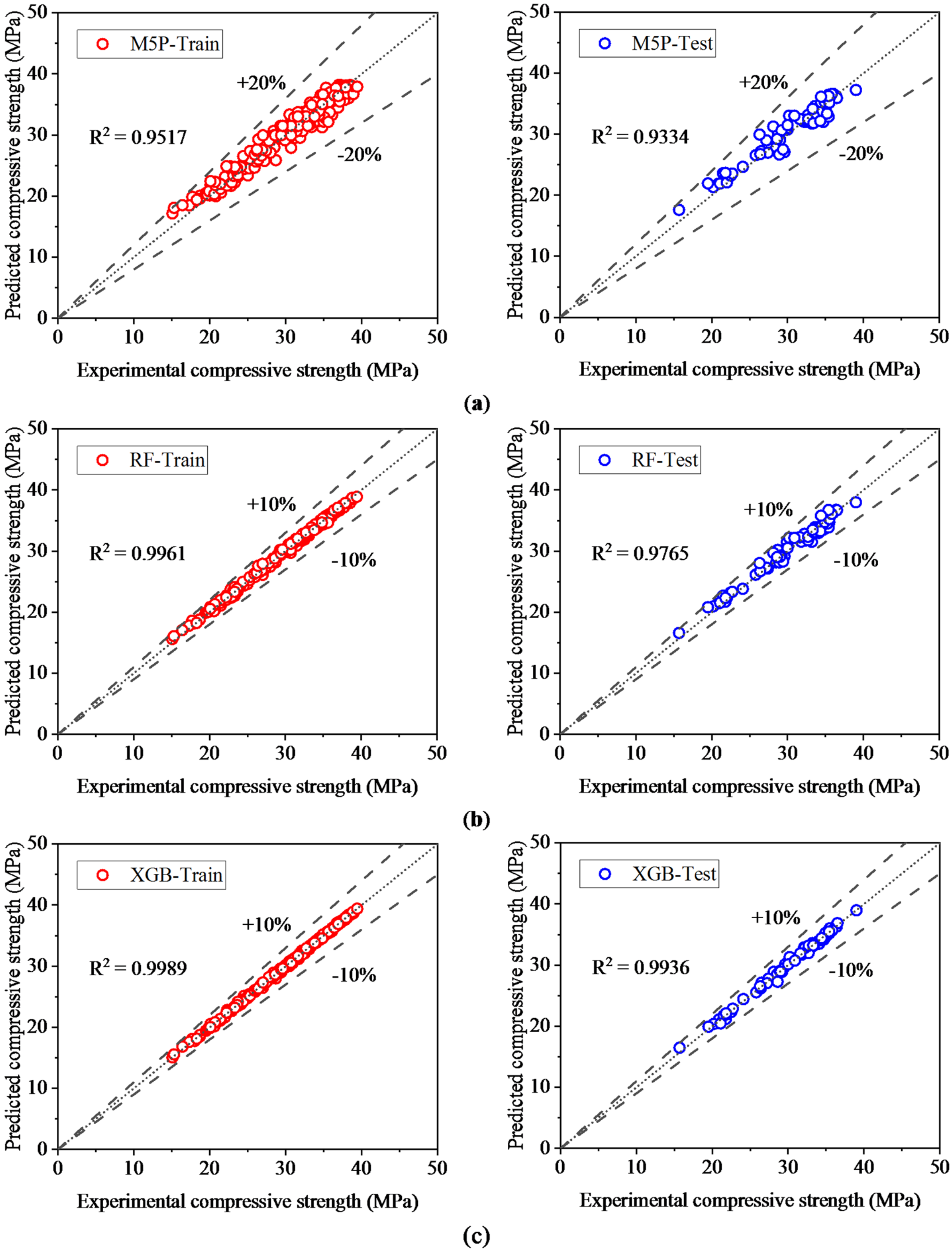


Fig. 13. Models' performance (**a**) M5P; (**b**) RF; (**c**) XGB.

strength prediction errors. Compressive strength scatter graphs compare experimental data against anticipated results. The train dataset scatter plot clusters around the optimal line, confirming that the predictions match the experimental results. The scatter plot of the test dataset shows points congregating near the ideal line, but it displays a slightly greater dispersion compared to the training data.

XGBoost model

Figure 12 presents the performance indicators of both actual values and compressive strength predictions utilizing XGB model. The scatter plots illustrate that all predicted values fall within the ± 10% error margins. The scatter plots of the training and testing datasets indicate that the points are strictly clustered around the optimal line, demonstrating that predictions from XGB model were closely matched with the test results. The XGB model demonstrates high R^2 and minimal prediction errors. The R^2MAPE, MAE, and RMSE values were 0.9989, 0.48%, 0.1351 MPa, and 0.1842 MPa for the train dataset, and 0.9936, 1.16%, 0.3278 MPa, and 0.4201 MPa for test dataset, respectively. Figure 12 shows that the XGB model predicts concrete compressive strength more accurately than the RF and M5P models due to its higher R^2 values and lower prediction errors. Figure 13c compares experimental and predicted compressive strength data from the model. The XGB model exhibited reliable accuracy in compressive strength predictions of concrete containing GWP and MWP under the effect of various temperatures.

Figure 14 presents the comparison between predicted and actual compressive strength for both training and testing datasets for all constructed prediction models. The RF and XGB algorithms demonstrated superior

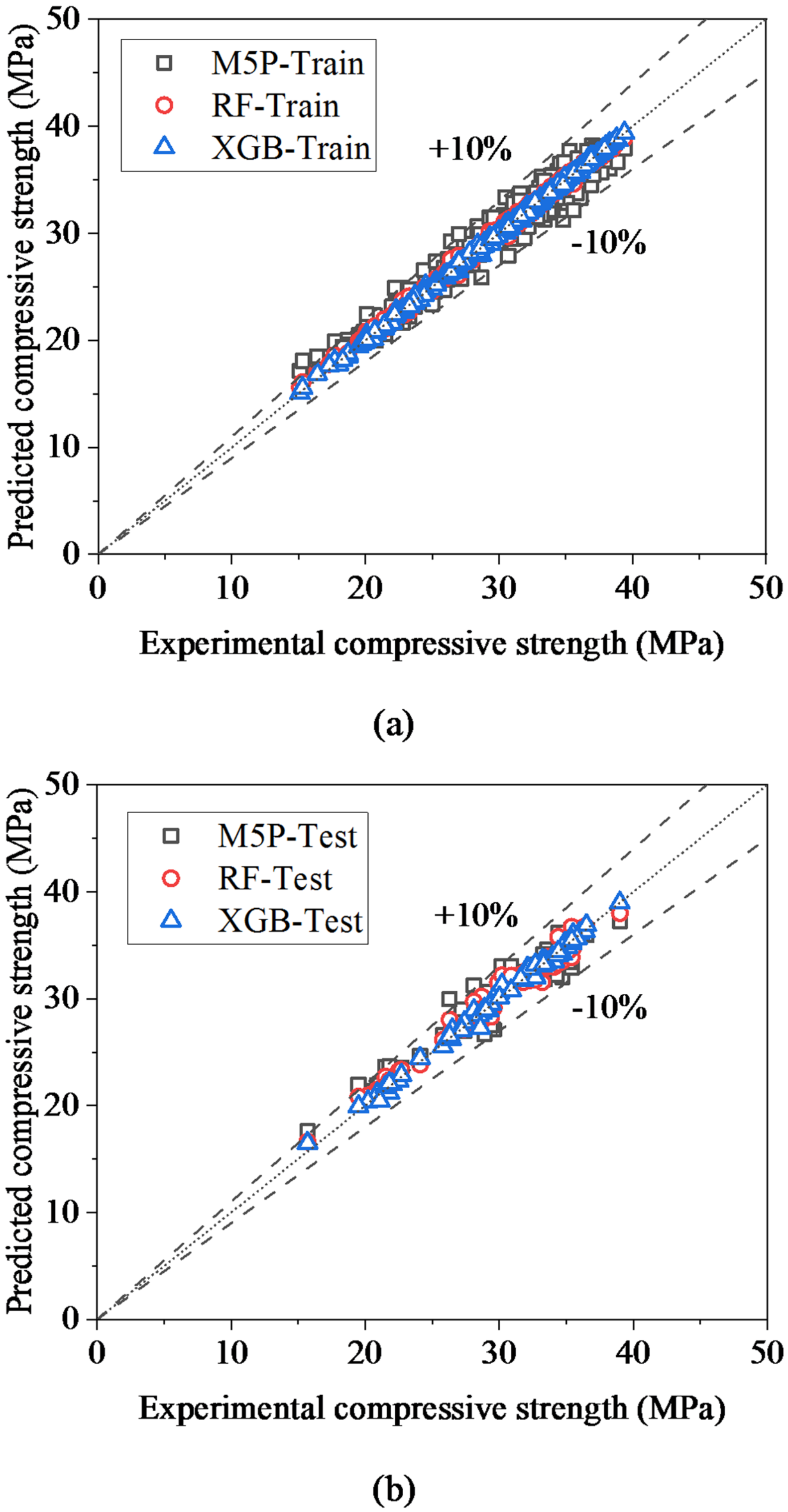


Fig. 14. Experimental values vs. predictions.

predictive performance relative to M5P model. The XGB model demonstrated higher accuracy, as the prediction results closely aligned with the experimental outcomes, indicated by the minimal scattering observed along the ideal line. The XGB model outperforms the M5P and RF models in predicting compressive strength, with higher R^2 values and lower error values, closely aligning with experimental results. The M5P model was less effective than the RF model. RF and XGB models outperform the M5P model in some scenarios, notably when data has complicated relationships or nonlinearities. Linearity assumptions and modeling interaction constraints greatly reduce M5P's performance. The models were ranked M5P, RF, and XGB, with XGB performing best.

Residual errors

Figure 15 presents a comparison between the residual errors in predicted compressive strength across all the proposed models. The XGB model demonstrated lower errors in predictions relative to RF and M5P models, indicating its superiority in compressive strength prediction of concrete modified GWP and MWP when subjected to high temperatures.

Analysis of K-folds

The analysis of k-folds was conducted to assess the outcomes of each algorithm (Fig. 16). This analysis is employed to assess the strength of the models to various datasets. The statistical measures were computed for the testing data to evaluate the efficiency of the proposed models across varying datasets. Figure 16 presents the ten K-fold results for all the constructed compressive strength models. The main purpose is to mitigate overfitting and biases in predictions. This technique is widely utilized due to its straightforward nature and its capacity to provide a more realistic assessment of model efficiency. After portioning the dataset into ten equal parts, a test dataset was derived from the ten folds and the remaining nine folds was the train dataset. After ten trials, scores improved significantly while maintaining precision.

The M5P model had R^2 values from 0.9125 to 0.9779, averaging 0.9441. R^2 values for the RF model ranged from 0.9633 to 0.9875, averaging 0.9792. The XGB model had R^2 values from 0.9776 to 0.9962, averaging 0.9884 (Fig. 16a). The values of MAE for M5P model ranged between 0.92 and 1.21 MPa, averaging 1.03 MPa (Fig. 16). MAE's RF-model values ranged from 0.504 to 0.741 MPa, averaging 0.63 MPa. XGB model MAE values ranged from 0.157 to 0.296 MPa, averaging 0.233 MPa (Fig. 16 b). RMSE for the M5P model ranged between 1.14 and 1.499 MPa, averaging 1.29 MPa. The RMSE of RF model ranged between 0.66 and 0.93 MPa, averaging 0.799 MPa. The XGB model's RMSE ranged from 0.219 to 0.487 MPa, averaging 0.328 MPa (Fig. 16c). The M5P model had a mean MAPE of 3.624% and ranged from 3.05 to 4.4%. MAPE for the RF model ranged from 1.74 to 2.89%, averaging 2.23%. The XGB model's MAPE ranged from 0.48 to 1.27%, averaging 0.83% (Fig. 16(d)). It can be concluded that the developed models can accurately predict unseen data in the range considered.

Feature importance

Figure 17 presents the feature importance plot, which illustrates the relevance of features in compressive strength prediction. The analysis indicates that feature T is the most significant across all models, followed by GWP and MWP. The analysis indicates that feature D is consistently the least significant across all models. The analysis reveals that the XGB and RF models attribute nearly equal weights.

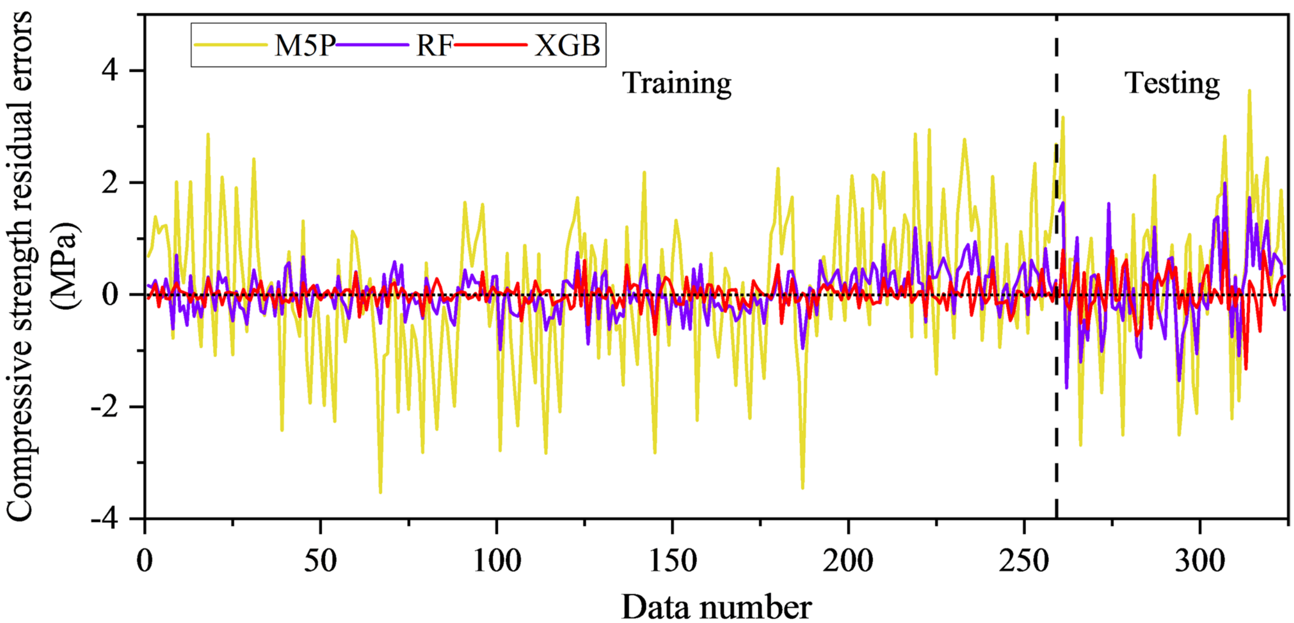


Fig. 15. Comparison between residual errors in prediction from all the proposed models.

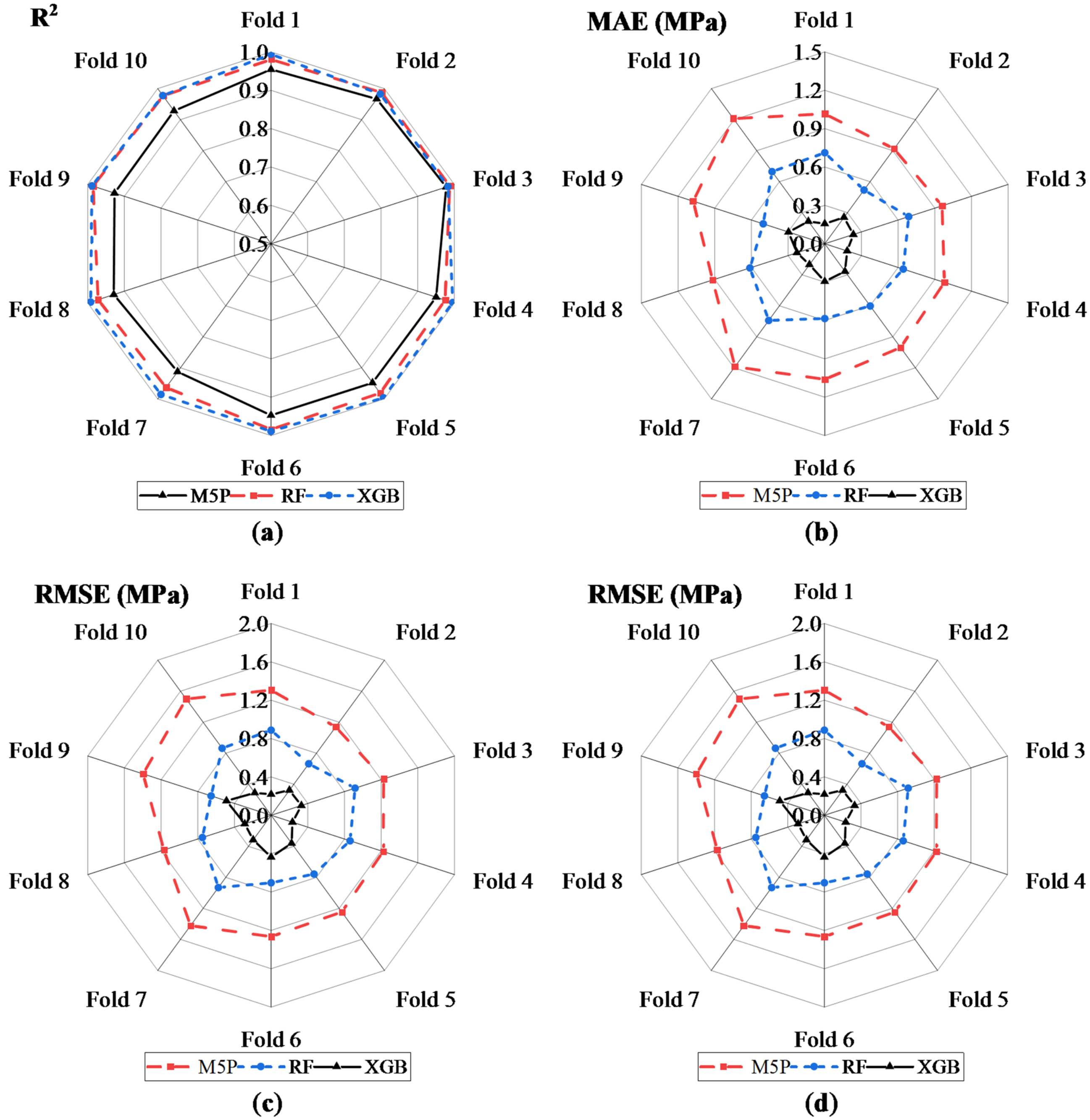


Fig. 16. Plots of the statistical measures of K-folds: (**a**) R^2; (**b**) MAE; (**c**) RMSE; (**d**) MAPE.

ML models enhanced explainability

Analysis of the utilized machine learning approaches and the dependencies of the input features was employed. To analyze the impact of each input parameter on the predictions, SHAP analysis was employed. The concrete compressive strength predictions for concrete containing GWP and MWP subjected to elevated temperatures were more precise when employing the RF and XGB algorithms; thus, these algorithms were selected for the SHAP analysis. Figure 18 illustrates the association of various features with SHAP values for RF and XGB models for the compressive strength prediction. Based on SHAP analysis, the temperature exhibited the high impact parameter affecting the compressive strength predictions, indicating that the temperatures highly affected compressive strength predictions. Concrete compressive strength generally decreases with increasing temperature. GWP positively influences compressive strength. The compressive strength can be enhanced as the dosage of GWP increases. Prolonged exposure to elevated temperatures commonly led to decreased compressive strength. D exhibits the least impact on compressive strength. MWP positively influences compressive strength, though its effect is less significant than that of GWP.

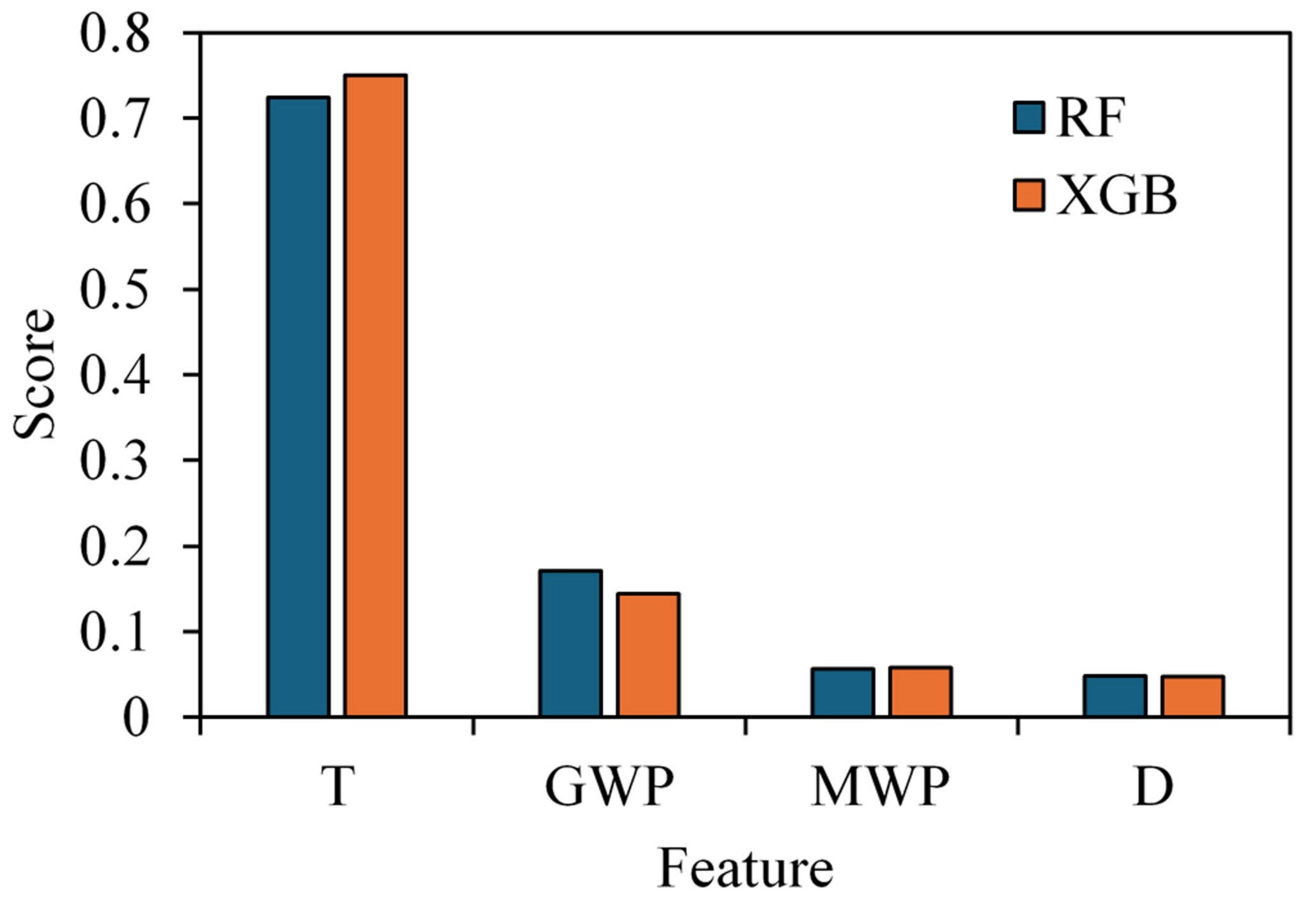


Fig. 17. Feature importance in the developed RF and XGB models.

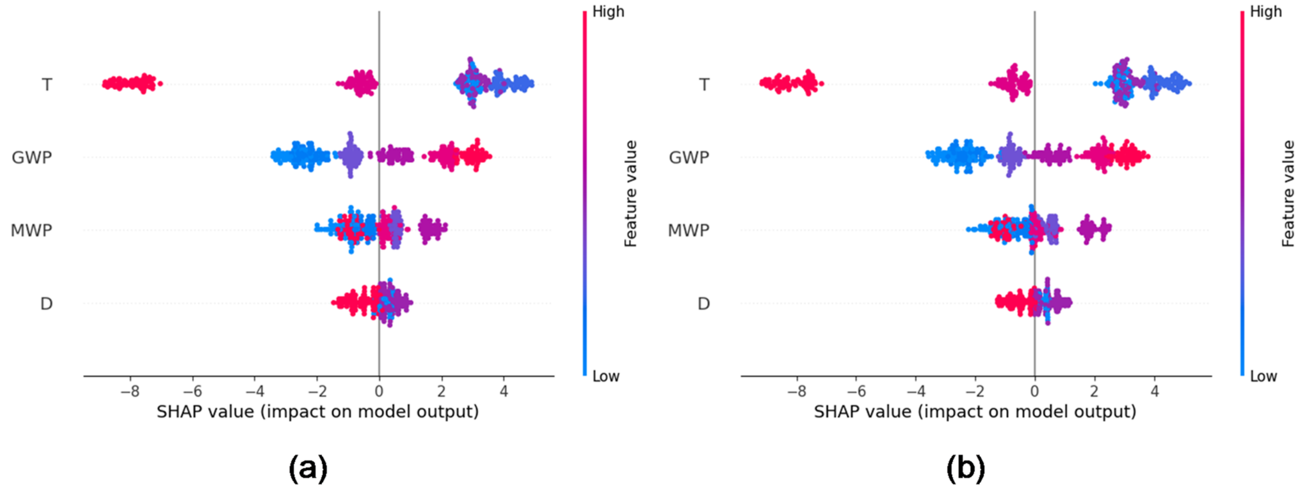


Fig. 18. SHAP plots: (**a**) RF-model; (**b**) XGB-model.

ICE with PDP

Figure 19 illustrates the impact of variations in each feature's value on the output result. The blue lines represent output estimations for each observation, while the dashed red line indicates the average of these estimations. The ICE plots demonstrate that compressive strength increases with elevated GWP values (Fig. 19a). The compressive strength (CS) rises with the percentage of MWP until it reaches 5%; beyond this point, the CS declines (Fig. 19b). The ICE plots demonstrate that compressive strength rises with increasing temperatures up to 200 °C, beyond which it decreases (Fig. 19c). The CS diminishes as exposure duration increases (Fig. 19(d)).

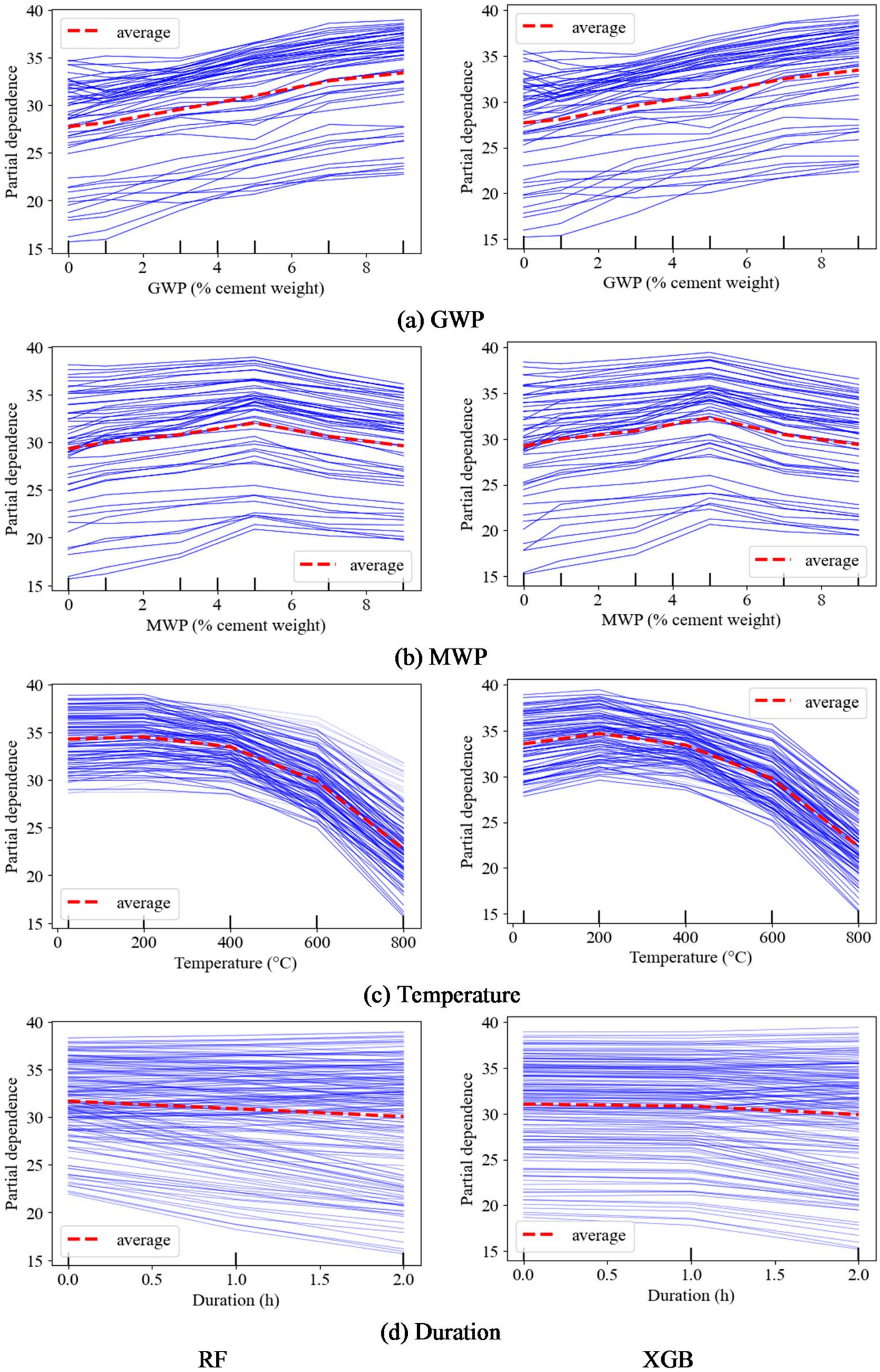


Fig. 19. ICE for input parameters.

The addition of GWP results in a significant increase in compressive strength, as demonstrated by the average line. Figure 20 presents the PDP, which demonstrates the average trend of ICE plots, aligning with the dashed red line depicted in the ICE plot. The variation in compressive strength in PDP is clearly evident. The horizontal axis represents the range of features, while the vertical axis signifies the predicted compressive strength. Increasing the GWP from 0 to 9% results in an increase in compressive strength from 27.7 MPa to 33.45 MPa (Fig. 20a). In a similar manner, incorporating MWP at a concentration of 5% enhances the compressive strength from 29.2 to 29.4 MPa (Fig. 20b). Elevated temperatures up to 400 °C do not significantly impact compressive strength;

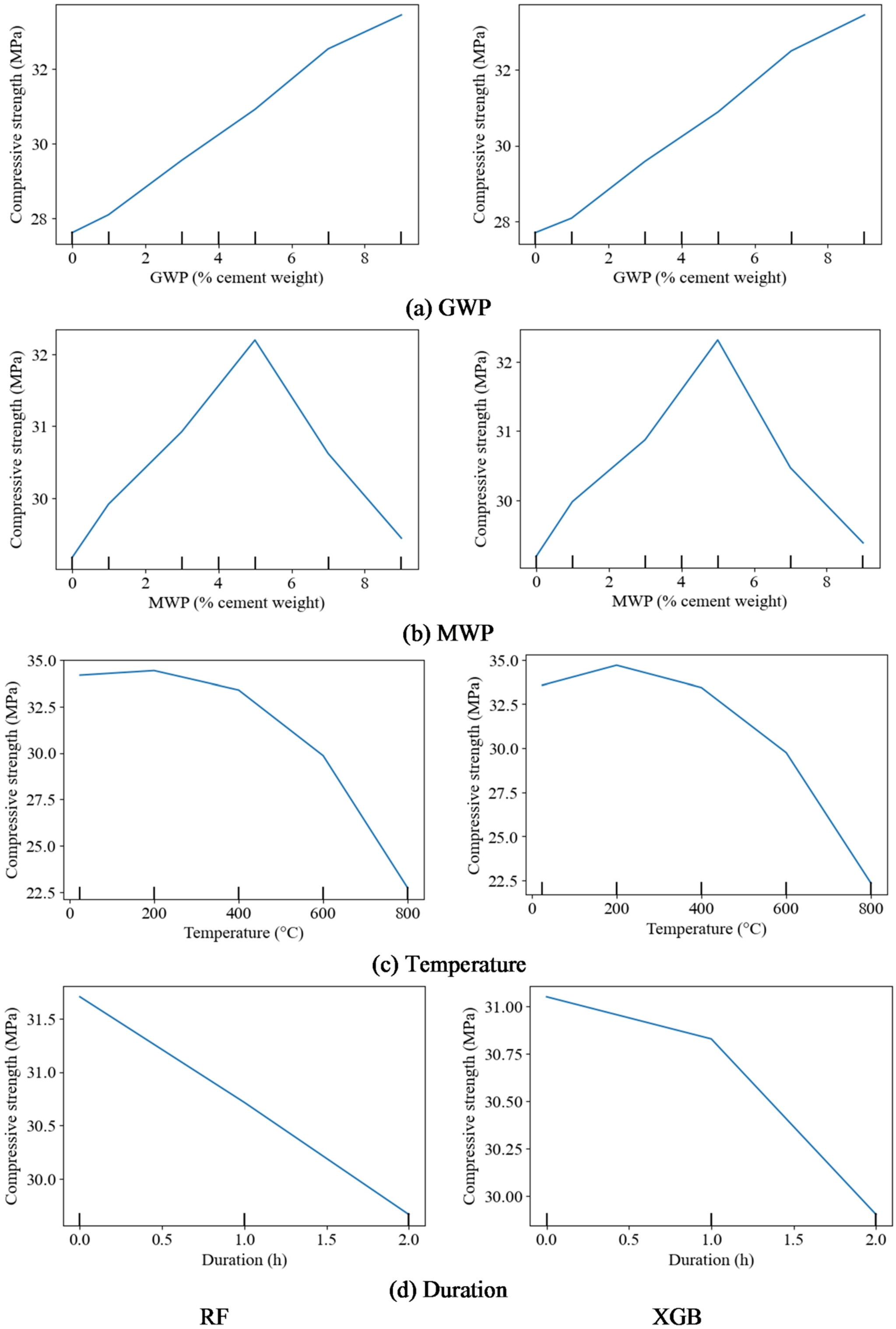


Fig. 20. PDP for input parameters.

however, increasing the temperature to 800 °C results in a reduction of compressive strength from 33.4 MPa to 22.8 MPa (Fig. 20c). The extension of exposure duration to 2 h decreases the compressive strength from 31 MPa to 29.7 MPa (Fig. 20(d)). GWP significantly enhances the compressive strength of concrete, whereas temperature adversely affects it to the greatest extent. Figure 21 illustrates the 2D-PDP, corresponding to the specified input parameters. Figure 21a indicates that the maximum CS is attained with GWP values ranging from 6.6 to 9% and MWP content between 1.57% and 6.3%. A maximum compressive strength of 33.69 MPa can be attained within these value ranges. Figure 21b indicates that the maximum CS is attained with GWP values ranging from

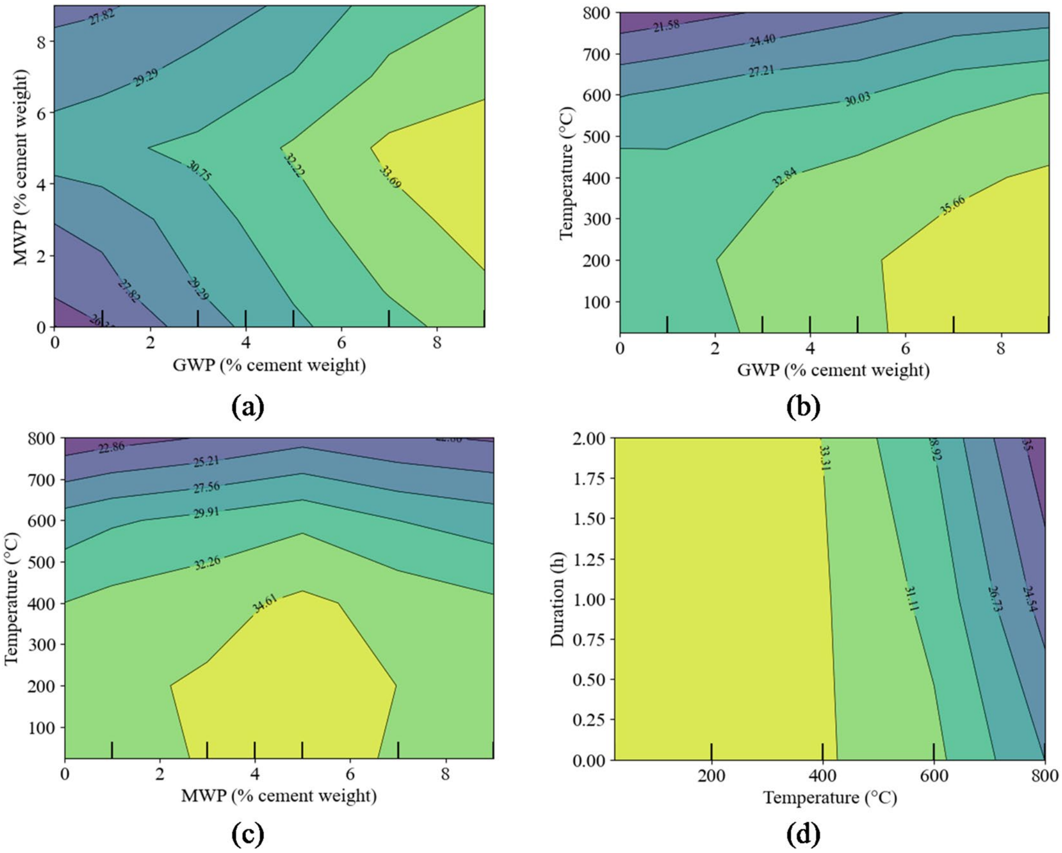


Fig. 21. 2D-PDP for input parameters.

5.6 to 9% and a temperature of up to 414 °C. A maximum compressive strength of 35.66 MPa can be attained within these value ranges. Figure 21c illustrates the combined influence of MWP and temperature on the CS. The compressive strength (CS) attains a peak value of 34.61 MPa when the moisture weight% (MWP) is between 2.63% and 6.6% and the temperature is below 417 °C. Figure 21d indicates that the maximum compressive strength of 33.31 MPa is attained at a temperature of 412 °C with exposure durations of up to 2 h.

Comparisons between earlier models and all the proposed models
Mahmoud et al.[8] developed water cycle algorithm (WCA), fuzzy logic (FL), genetic algorithms (GA), artificial neural networks (ANN), and multiple linear regression (MLR) for predicting post-heating compressive strength of concrete incorporating GWP and MWP. The comparison was made utilizing various statistical measures (Fig. 22). The XGB model exhibits enhanced performance relative to other compressive strength models, showcasing the peak R^2 values while concurrently achieving the lowest RMSE and MAE metrics.

Graphical user interface
A graphical interface has been developed utilizing Python framework to help in the prediction of compressive strength of concrete containing GWP and MWP under various temperatures, utilizing RF and XGB models. The graphical user interface is organized according to four primary key variables: GWP (percentage of cement weight), MWP (percentage of cement weight), temperature (°C), and exposure duration (h). This tool is efficient, conserving time and resources by minimizing the necessity for physical experiments while providing rapid and precise results that can greatly improve efficiency in research and industry contexts. Table 6 presents the details of the two models, while Fig. 23 illustrates the results of the GUI calculations. The comparison of experimental and predicted values indicates that the results obtained through the GUI are satisfactory.

Conclusions
This study developed prediction models for compressive strength of concrete incorporating GWP and MWP when subjected to high temperatures, utilizing three machine learning approaches: M5P, RF, and XGB. The following conclusions are drawn:

1. The incorporation of GWP and MWP improved the concrete compressive strength and mitigated the impact of elevated temperatures.
2. The developed prediction models M5P, RF, and XGB can be utilized for predicting compressive strength of concrete containing GWP and MWP as cement partial replacements.

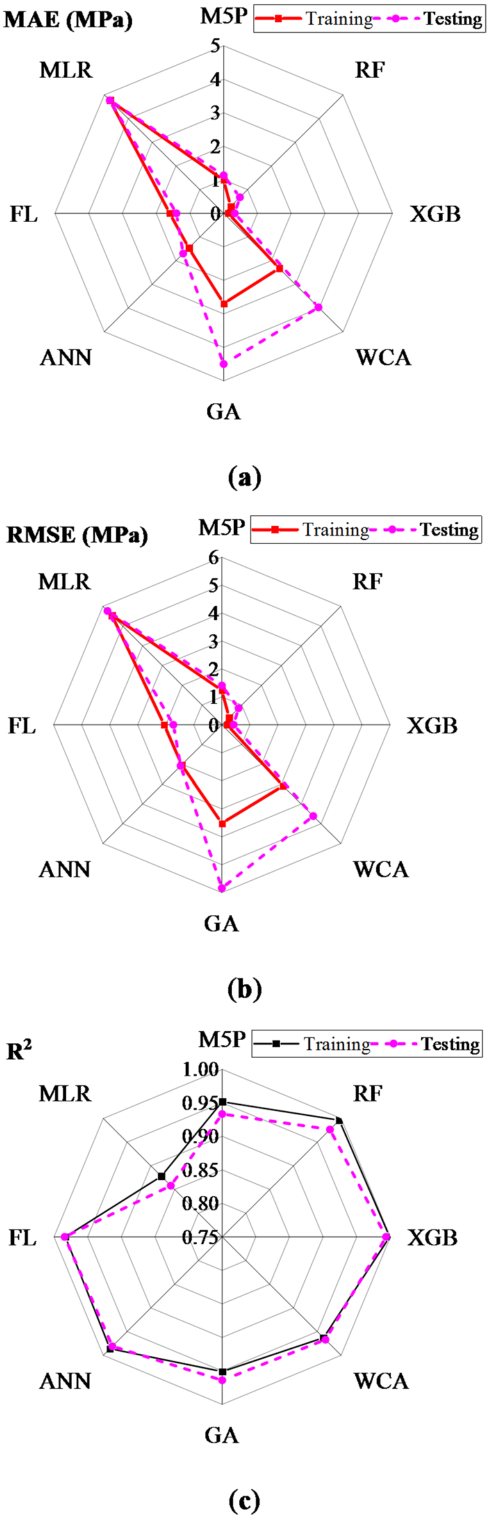


Fig. 22. Comparison between the proposed models (M5P, RF, and XGB) and earlier developed models (WCA, GA, ANN, FL, and MLR).

3. The XGB technique demonstrated increased efficiency, evidenced by R^2 values of (0.9989, 0.9936) for the training and testing sets. It also showed the lowermost MAE of (0.1351 MPa, 0.3278 MPa), along with RMSE of (0.1842 MPa, 0.4201 MPa), and MAPE values of (0.48%, 1.16%). The XGB model demonstrates enhanced comprehensiveness and suitability for predicting concrete's compressive strength when incorporating GWP and MWP, particularly under high-temperature conditions.

GWP (%)	MWP (%)	Temperature (°C)	Duration (h)	Compressive strength (Actual/RF-model/XGB-model)
1	3	25	0	31.1/31.16/31.11
5	3	25	0	35.7/35.74/35.7
3	5	200	1	34.9/34.71/34.97
7	5	600	1	35/34.55/35.12
1	5	200	2	35.4/34.87/35.53
5	7	600	1	29.9/30.28/29.85
3	3	600	2	29/29.03/29.02
5	5	800	2	22.9/22.37/22.82

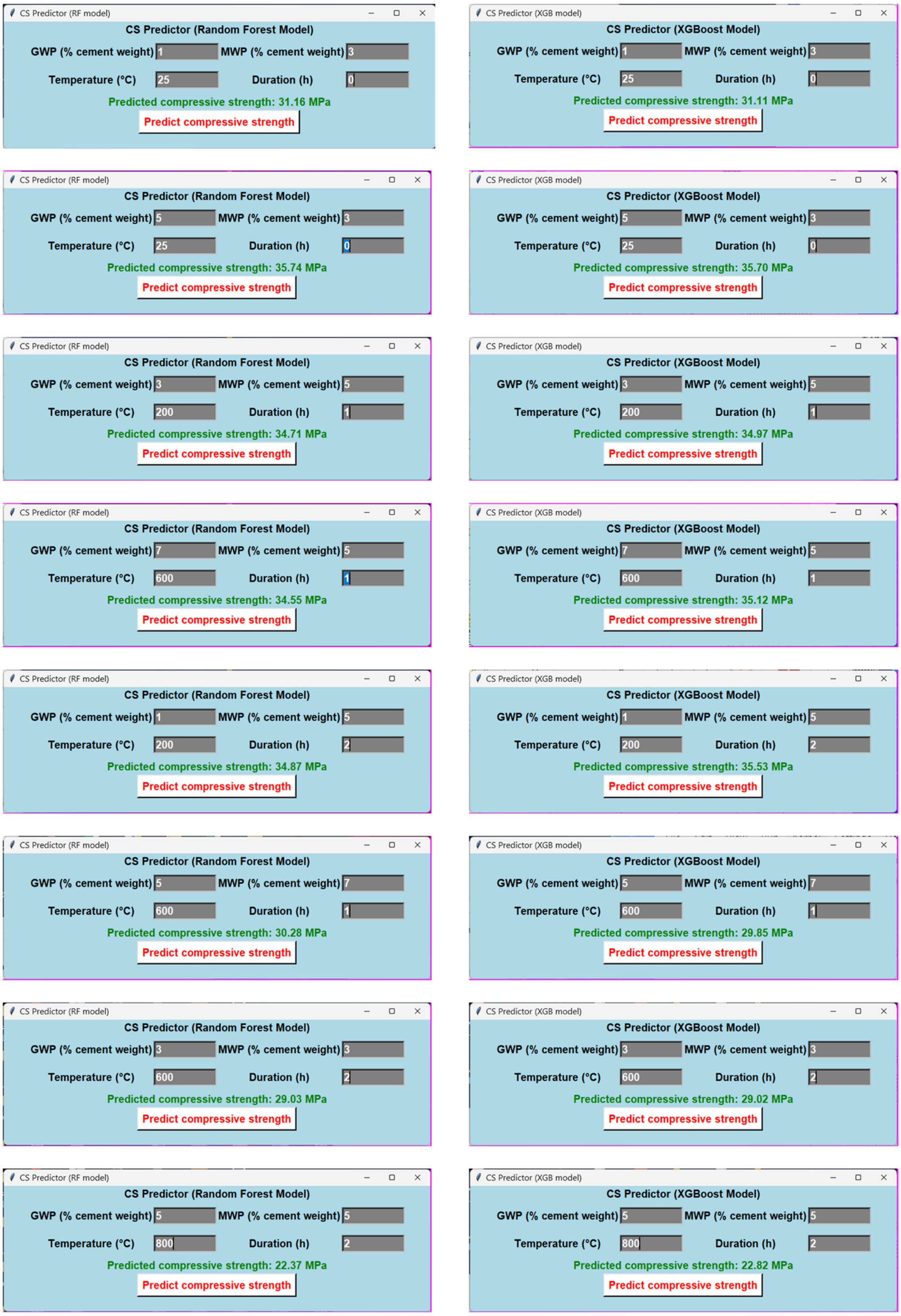
Table 6. Details of eight scenarios.

4. The validation by K-fold indicated the enhanced efficiency and accuracy of all the proposed models. The SHAP, ICE, and PDP analyses indicate that the temperature exerts the most significant negative influence on model performance in predicting the compressive strength. The GWP and MWP demonstrate a positive influence on the prediction of the compressive strength (CS), whereas the variable D exhibits the least influence.
5. In comparison with earlier models for compressive strength predictions of concrete incorporating GWP and MWP, the performance of established the XGB model out performed other models: WCA, GA, ANN, FL, and MLR.
6. A graphical user interface has been created to accurately forecast the compressive strength of concrete containing GWP and MWP exposed to elevated temperatures using essential input variables, reducing the need for resources and experimental tests.

Limitation, future works, and possible real-world scenarios

This study utilized various machine learning techniques to forecast the compressive strength of concrete, whereby granite and marble waste powders partially substituted cement at amounts between 1% and 9%. Concrete specimens were subjected to temperatures reaching 800 °C for periods ranging from one to two hours. The study included four variables: waste material type, waste material %, temperature level, and exposure period, yielding a dataset of 324 residual compressive strength values; nevertheless, the findings are fundamentally constrained by the limitations of the experimental data. Therefore, the findings are not applicable to various cement replacement ratios, different types of waste-based concrete, or varying temperature exposure conditions. This limitation, however, offers prospects for future research to improve the relevance and influence of these preliminary findings. Future studies should consider a broader range of cement-replacing materials, concrete ingredients, rate of temperature effects, and environmental factors to enhance the models' generalizability. Exploring alternative machine learning techniques, including hybrid learning models, may enhance predictive skills. Looking into how practical and cost-effective it is to use GWP and MWP in making concrete would give a better idea of how they could be used on a large scale.

The findings of the current study hold substantial real-world application potential, particularly in promoting sustainability and environmental preservation when implemented on a large scale. Notably, in Egypt's Shaq El-Teeban area, agreements have already been reached for the establishment of factories that will utilize vast quantities of marble and granite waste, repurposing them into products like curbstones and cement bricks. Furthermore, ongoing research is investigating the feasibility of incorporating these materials into pozzolanic cement production. In addition, the study contributes to circular economy initiatives by providing evidence for the effective reuse of industrial waste in enhancing concrete performance, thus reducing environmental impact and promoting resource efficiency. On the other side, the developed ML models can be applied in post-fire structural assessments to estimate the residual compressive strength of concrete, thereby aiding engineers in evaluating safety and determining appropriate rehabilitation strategies. Furthermore, the demonstrated use of granite and marble waste powders supports green building practices and aligns with sustainability-oriented construction codes that encourage the incorporation of recycled materials. The findings also inform the design of fire-resilient infrastructure in high-risk areas, where the thermal performance of concrete is critical.



(a) RF model (b) XGB model

Fig. 23. GUI environment for predicting CS of MSC.

Data availability

Raw data supporting the conclusions of this article will be made available by the corresponding author upon reasonable request.

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Declarations

Competing interests
The authors declare no competing interests.

Additional information
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